

Research Briefing

20 June 2025

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Electric Vehicles and Infrastructure



Summary

- 1 Progress on the electric vehicle transition
- 2 Government policy since 2020
- 3 The integration of EVs into the energy system
- 4 Environmental impact: EV lifecycle emissions and battery decommissioning
- 5 Past government EV policy pre-2020

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Summary

The UK has [committed to net-zero carbon emissions](#) by 2050. [Transport was responsible for 29% of total UK greenhouse gas emissions](#) in 2023, making it the largest emitting sector of the UK economy. Over half (54%) of the [UK's transport emissions](#) come from cars and taxis.

Electric vehicles offer a way to reduce these emissions, because full battery electric or hydrogen fuel cell vehicles produce zero carbon emissions at the tailpipe. In February 2024 [the government said](#) “the transition to electric vehicles (EVs) is a crucial step towards achieving the UK's net zero target.”

In April 2025 the government [confirmed an end to the sale of new pure petrol or diesel cars in 2030](#), which was a [manifesto commitment](#). It also set an end to new pure petrol-diesel vans in 2035, and that the sale of hybrid cars would be permitted between 2030 and 2035.

In 2021 the [government said that 2040](#) would be a ‘backstop’ phase-out date for new non zero-emission vehicles of all other types, including HGVs and buses.

Progress on the EV transition

The market for EVs is small yet growing. Just [3% of cars \(1.09 million\) were battery electric](#) in the UK at the end of June 2024, and 8% were hybrid electric (using an electric motor as well as petrol/diesel). Battery electric cars accounted for [19% of all new car registrations](#) at the end of June 2024, an increase from 1% in 2018.

In its 2024 inquiry into EV strategy, the [House of Lords Environment and Climate Change Committee said](#) that demand for electric cars is being constrained due to their upfront cost, inadequate charging infrastructure and general consumer scepticism.

Zero Emission Vehicle (ZEV) Mandate

In January 2024, the government introduced a [zero emission vehicle \(ZEV\) mandate](#) for car manufacturers, intended to provide greater certainty to manufacturers, and provide a greater range of EV options to consumers.

The [mandate](#) specifies the minimum proportion of car manufacturers' sales that must be zero-emission vehicles. This will increase from 22% in 2024 to 80% by 2030, and 100% in 2035. Following a consultation, in April 2025 the government said it will relax [some of the ZEV mandate rules](#).

Improving charging infrastructure

There are a variety of UK-wide [grant schemes to support new EV chargepoints](#) outside homes and at workplaces. In England, [Local Electric Vehicle Infrastructure \(LEVI\)](#) funding is available for local authorities. In 2020 the Conservative government also announced plans for a [Rapid Charging fund \(RCF\)](#) to increase provision of chargepoints at motorway service stations, but the RCF has still not yet opened to bids.

In 2021 the government [legislated to require EV chargepoints to be installed](#) in a range of new build homes and other buildings. In 2023 it [introduced regulations](#) to improve the [customer experience of on-street EV charging](#), including contactless payments. In 2022 the British Standards Institute (BSI) produced a [voluntary specification for the installation of accessible chargepoints](#), to improve EV charging for people with disabilities.

In December 2024 the government published guidance for local authorities to help them develop policies on [cross-pavement charging solutions](#).

The devolved administrations in [Scotland](#), [Wales](#) and [Northern Ireland](#) also have their own strategies for EV infrastructure.

EVs and electricity demand

Increasing the number of EVs will reduce petrol and diesel demand but add to electricity demand. This could increase pressure on Great Britain's electricity grid network, operated by [National Grid Electricity System Operator](#) (ESO). There has previously been concern about the challenges to the grid, in terms of electrical capacity and grid balancing.

However National Grid ESO has said that although demand for electricity will increase, [there will be a managed transition that will be within the range the grid can handle](#).

EVs also have the potential to help balance a more flexible and low carbon grid through developments that allow their batteries to charge at optimum times, [known as "smart charging", and to supply electricity in times of high demand, known as "vehicle to grid"](#).

EVs and environmental impact

The total greenhouse gas emissions from an EV are known as its “lifecycle emissions”. These combine the emissions from manufacturing the vehicle (which tend to be higher than manufacturing internal-combustion vehicles), powering the vehicle through its life (which tend to be lower than powering internal-combustion vehicles), and decommissioning the batteries at the end of their life.

Calculations of lifecycle emissions vary based on the methodology and assumptions used. The International Energy Agency’s [lifecycle emission comparisons](#) calculates that in the UK, a medium sized EV has higher manufacturing emissions than an internal-combustion engine equivalent, but the lower emissions from powering the EV mean its cumulative emissions are lower after 3 years of use.

There are various other environmental considerations related to EVs, including the use of [critical minerals](#) in batteries, which have challenges with sustainable extraction, and how best to recycle or reuse batteries.

1

Progress on the electric vehicle transition

Electric vehicles (EVs) could help the UK decarbonise its transport sector and meet its net zero commitments. The number of electric cars on the roads is growing, while progress has been slower for electric vans and lorries. A range of barriers to demand persist include their upfront cost, access to chargepoints, and general consumer scepticism.

1.1

Why do we need electric vehicles?

Benefits

Decarbonisation potential

In 2021 the government said that removing tailpipe emissions from cars and vans is ‘fundamental’ to decarbonising transport and meeting its wider commitment to reach net zero emissions by 2050.¹

Similarly, in 2024 the Labour government said:

The ZEV [zero emission vehicle] transition is ... critical for tackling climate change. For the 11th consecutive year, domestic transport is the largest emitting sector of our economy... and, as such, it must be front and centre of our efforts to accelerate to net zero and to tackle the global climate crisis.²

Transport produced 29% of the UK’s total greenhouse gas (GHG) emissions in 2023 and remains the largest emitting sector in the UK.³

The vast majority (89%) of domestic UK transport emissions came from road vehicles in 2023. The biggest contributors were cars and taxis, which made up just over half (54%) of all domestic transport emissions in 2023 (see graph below).⁴ Moving to fully electric vehicles (EVs) could help reduce GHG emissions as well as eliminating other tailpipe pollutants such as nitrogen dioxide, and cutting vehicle noise.⁵

¹ DfT, [Decarbonising transport: Greener, Better Britain](#), July 2021, p 88; [Section 1 of the Climate Change Act 2008](#)

² DfT Consultation, [Phasing out sales of new petrol and diesel cars from 2030 and supporting the ZEV transition](#), 24 December 2024, para 4

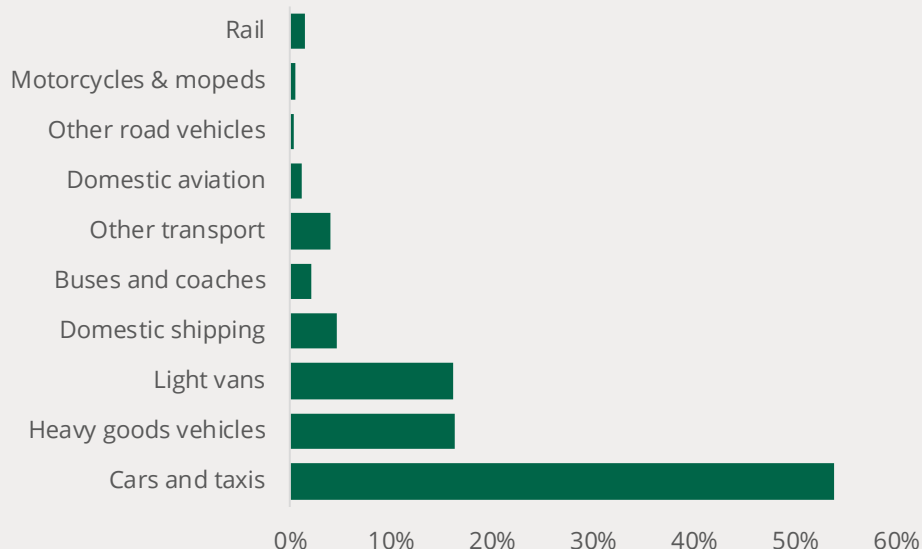
³ (DfT), [Energy and environment: data tables](#), ENV0201

⁴ (DfT), [Energy and environment: data tables](#), ENV0201

⁵ Local Government Association, [The case for electric vehicles](#) [accessed 7 June 2024]

UK transport greenhouse gas emissions, 2023

Percentage of greenhouse gas emissions by transport mode



Notes:

- 'Other road transport emissions' consist of emissions from road vehicles running on liquified petroleum gas (propane and butane), emissions from the evaporation of engine lubricants and urea as well as urea use, and biofuel use.
- There are no direct emissions from electric trains. However the end user emissions do include the emissions resulting from the production of the electricity used by electric trains.
- Other mainly consists of 'military aircraft and shipping' & 'aircraft support vehicles'.
- International aviation and international shipping emissions are not included

Source: Source: Department for Transport (DfT), [Energy and environment: data tables](#), ENV0201

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- International aviation and international shipping emissions are not included

Lower running costs

2023 analysis by the Energy & Climate Intelligence Unit, a non-profit climate organisation, found that compared to an ICE equivalent, an EV can cost up to £700 less to run per year.⁶ The government has said that switching to EVs

⁶ Energy & Climate Intelligence Unit, [New analysis: petrol car drivers paid a £700 'petrol premium' in 2023](#), 30 December 2023

may therefore directly benefit private car owners as well as being financially “transformative” for commercial fleets, freight, logistics, bus, and taxi industries.⁷

Limitations

Non-exhaust pollution and reliance on cars

The EU’s European Environment Agency has said that while EVs offer “clear benefits” for local air quality due to zero exhaust emissions at street level, they still emit particulate matter from road, tyre and brake wear. This means electric vehicles cannot entirely eliminate transport-based air pollution in cities.⁸

Further, EVs do not address wider issues of urban sprawl, inactive lifestyles or congestion, which could increase if the transition to EVs means driving becomes cheaper.⁹ Both the Centre for Research into Energy Demand Solutions (CREDS) and the Climate Change Committee (CCC) – an independent, statutory body which advises the Government on decarbonisation – have warned against a reliance on EVs and recommend the Government should also reduce demand for cars more generally.¹⁰

Slow transition and need to decarbonise the grid

In 2021 the government acknowledged that the transition to EVs may be slow and in 2035 over half of the car fleet might still not be zero-emission.¹¹ In addition, until the electricity grid is fully decarbonised, any carbon savings from EVs might be partially offset if people drive them more.¹²

Tax revenues

As noted in the Transport Select Committee’s 2022 report into road pricing, the transition to EVs may mean tax revenues from vehicle excise duty, and in particular fuel duty, will decline. The committee recommended the government start work on a national road pricing system, based on miles travelled and vehicle type, which would enable the Government to maintain the existing link between motoring taxation and road usage and avoid a ‘fiscal black hole’.¹³ HM Treasury responded to the report on 9 January 2023,

⁷ DfT, [Decarbonising transport: Greener, Better Britain](#), July 2021, p28

⁸ European Environment Agency, [Electric vehicles from life cycle and circular economy perspectives](#), 22 Nov 2018

⁹ Centre for Research into Energy Demand Solutions, [Shifting the focus: energy demand in a net-zero carbon UK](#), 5 July 2019

¹⁰ CCC, [2023 Progress Report to Parliament](#), 28 June 2023, p128

¹¹ DfT, [Decarbonising transport: Greener, Better Britain](#), July 2021, p29

¹² DfT, [Decarbonising transport: Greener, Better Britain](#), July 2021, p29

¹³ Transport Committee, [Road pricing](#), 4 February 2022, HC 789, Recommendation 10; Transport Committee Press Release, [Road pricing: Act now to avoid £35 billion fiscal black hole, urge MPs](#), 4 February 2022

saying “the government did not currently have plans to consider road pricing.”¹⁴

Manufacturing emissions

The government commissioned a lifecycle analysis of UK road vehicles by the consultancy firm Ricardo, which was published in 2022. This analysis concluded that EVs generally have higher manufacturing emissions than ICE (that is fully petrol/diesel) vehicles but that, crucially, EVs have far lower emissions than ICE vehicles over their whole lifecycle.¹⁵

The Ricardo analysis said that in 2020 a typical battery electric car was estimated to save ~65% GHG emissions compared to an equivalent conventional petrol car.¹⁶ The analysis suggests EV lifecycle emission savings may increase further (to ~75% by 2030 and ~81% by 2050) due to future improvements in manufacturing processes, in vehicle and battery recycling, and in potential second-life applications for EV batteries.¹⁷

1.2

What are electric vehicles?

EVs derive some or all of their power from large, rechargeable batteries. Some also have a conventional fossil fuel engine as well as a battery. These are called hybrid vehicles. The distance an EV can drive between charges is known as its range.

Zero-emission vehicles

Zero-emission vehicles (ZEVs) have no tailpipe CO₂ emissions. There are two main types of ZEVs:

Battery electric vehicles

Battery electric vehicles (BEVs) are vehicles where the battery is the only power source. The current average quoted range of new BEVs is between 230 and 300 miles, according to the industry body, the Society of Motor Manufacturers and Traders (SMMT).¹⁸ In practice, range varies according to driving style, terrain and the use of auxiliary equipment such as heating/air conditioning. Examples of BEVs include the Tesla Model 3 and Nissan Leaf.¹⁹ Most zero-emission buses are also BEVs.

¹⁴ Transport Committee, [Letter to the Chair from Rt Hon Jeremy Hunt MP, Chancellor of the Exchequer](#), 9 January 2023

¹⁵ DfT, [Lifecycle analysis of UK road vehicles](#), 25 March 2022, p6

¹⁶ DfT, [Lifecycle analysis of UK road vehicles](#), 25 March 2022, ix

¹⁷ DfT, [Lifecycle analysis of UK road vehicles](#), 25 March 2022, p6

¹⁸ SMMT News, [Brits enjoy best ever EV choice with more than a hundred models now available](#), 23 May 2024

¹⁹ DfT, [Vehicle Licensing Statistics: 2021 Quarter 2 \(Apr - Jun\)](#) [PDF], 29 September 2021

²⁰ DfT, [Vehicle licensing statistics data tables](#), VEHO141b (GenModels)

BEV charging times can vary from as little as 15-20 minutes using a rapid charger, to up to 22 hours using a 3-pin plug at home.²¹

Fuel cell electric vehicles

Fuel cell electric vehicles (FCEVs) are sometimes referred to as hydrogen powered. These generate their own electricity on-board from a fuel such as hydrogen, and do not need to plug in to the electricity grid to recharge. There are currently very few FCEVs available to buy and very few re-fuelling stations open to the public in the UK.²² Examples of FCEVs include the Toyota Mirai and Hyundai IX35.²³

FCEVs have a similar range to petrol/diesel equivalents. Re-fuelling is similar to a petrol vehicle and usually takes less than ten minutes. FCEVs are currently more expensive than BEVs and petrol/diesel equivalents, both in terms of up-front cost and running costs.²⁴

Fuel cell technology is often seen as having potential for decarbonising long-distance HGVs, although this would rely on growing the production of low carbon hydrogen and creating dedicated infrastructure.²⁵

FCEVs currently have similar life-cycle emissions to BEVs, although FCEVs emissions could reduce dramatically in future if the production of low carbon hydrogen is scaled up and becomes more efficient.²⁶ Currently the UK only has about 5 MW of hydrogen production capacity.²⁷ The Government has plans to grow hydrogen production to 10 GW by 2030, but these are still at an early stage.²⁸

Hybrid vehicles

Hybrid vehicles have both a petrol or diesel combustion engine and an electric motor. They generate tailpipe emissions when the combustion engine is being used.

Plug-in hybrids

Plug-in hybrid electric vehicles (PHEVs) switch between running on their battery and on fossil fuels. They typically have a smaller battery than BEVs, and therefore a lower battery powered range of between 20 to 30 miles. This is enough to complete many short urban journeys on electric power whilst emitting zero tailpipe emissions. When the battery is depleted they can use their conventional engine, meaning their maximum range is equivalent to a

²¹ Pod Point, [How long does it take to charge an electric car](#), 14 June 2024

²² Driving Electric, [Where can I buy hydrogen and where is my nearest hydrogen filling station?](#), 11 April 2024

²³ DfT, [Vehicle Licensing Statistics: 2021 Quarter 2 \(Apr - Jun\)](#) [PDF], 29 September 2021

²⁴ RAC, [Hydrogen cars: are they the future?](#) 16 September 2020

²⁵ Engineering Update, [Fuel cell versus battery trucks](#), 11 April 2023

²⁶ DfT, [Lifecycle analysis of UK road vehicles](#), 25 March 2022, p6, Figure 2.1

²⁷ Renewables UK, [New guide shows how to scale up UK's green hydrogen industry](#), 1 November 2023

²⁸ National Grid, [Future Energy Scenarios 2023 Report](#), July 2023, p12

petrol or diesel car. Their batteries are recharged by plugging them in to the electricity grid.²⁹ Examples of PHEVs include the BMW X5 PHEV and Mitsubishi Outlander PHEV.³⁰

Hybrid electric vehicles

Hybrid electric vehicles (HEVs) do not plug in and have a much smaller battery which is recharged while driving. Hybrid electric vehicles can drive in electric mode for just a few miles and have higher emissions than PHEVs.³¹ Examples of HEVs include the Toyota Yaris HEV.³²

Classifying electric vehicles

Some of these types of electric vehicles qualify as zero-emission or ultra-low-emission vehicles, whereas others do not because their emissions are too high. Zero-emission vehicles emit no CO₂ emissions at the tailpipe. BEVs and FCEVs fall within this category.

Vehicles with reported tailpipe emissions of less than 75 g/km of CO₂ are classified by the government as 'ultra-low emission vehicles' (ULEV), leading to some tax breaks.³³ Some PHEVs models meet this criteria, most HEVs do not.³⁴

The following DfT table explains how the different types of EV compare in terms of electric fuel types.

²⁹ What Car? [What is a plug-in hybrid and should you buy one?](#), 11 September 2022

³⁰ DfT, [Vehicle Licensing Statistics: 2021 Quarter 2 \(Apr - Jun\)](#) [PDF], 29 September 2021

³¹ What Car? [Best self-charging hybrid cars you don't have to plug in 2024](#), 11 May 2023

³² DfT, [Vehicle Licensing Statistics: 2021 Quarter 2 \(Apr - Jun\)](#) [PDF], 29 September 2021

³³ DfT Guidance, [Ultra low emission vehicles: tax benefits](#), 24 May 2018

³⁴ DfT, [Vehicle Licensing Statistics: 2021 Quarter 2 \(Apr - Jun\)](#) [PDF], 29 September 2021

What electric fuel types are there?		Does the vehicle use petrol or diesel?	
		Yes	No
Does the vehicle use electric power? 	Yes, and is a plug-in	Plug-in hybrid electric (PHEV)*	Battery electric (BEV)
	Yes, but is not a plug-in	Hybrid electric (HEV)	Fuel cell electric (FCEV)
	No	Petrol / Diesel	Other**

* A range extended electric vehicle is a special case of PHEV, where the conventional fuel does not power the wheels directly, usually only charging the battery for additional range.

** This table excludes rare combinations based on biofuels and other emerging technologies.

DfT, [Vehicle licensing statistics: 2021](#), 24 May 2022

‘Technology neutral’ or ‘picking winners’?

Since 2018, the government has said that it is “technology neutral” regarding what technology or technologies may achieve the desired outcome of achieving net zero.³⁵ This principle also applies to zero emission vehicles. In 2023 the government provided the following definition which is wider than just battery EVs:

A zero emission vehicle is a vehicle that does not produce harmful exhaust emissions, such as greenhouse gas and pollutant emissions. Exhaust emissions that do not damage public health, such as water vapour, are permitted.

UK government policy on the transition to zero emissions is technology neutral.³⁶

In their 2023 inquiry ‘Fuelling the future’, the Transport Select Committee heard that technological neutrality has been a barrier to investment in decarbonisation technologies as it had created uncertainty about which

³⁵ DfT, [Government launches Road to Zero Strategy to lead the world in zero emission vehicle technology](#), 9 July 2018

³⁶ DfT, Closed call for evidence, [Infrastructure for zero emission heavy goods vehicles and coaches](#), 19 October 2023

technologies would be ‘winners’.³⁷ It recommended the government shift to a ‘targeted technology investment’ policy where it could invest in evidenced solutions to lowering emissions, while maintaining neutrality in sectors which are harder to decarbonise.³⁸

The previous government disagreed, saying that given the uncertainty of future technological advances, its strategy remained “the most credible pathway to deliver on its net zero targets” and would be reviewed periodically.³⁹

The current government has also said it remains “technology-neutral” when it comes to replacing ICE vehicles.⁴⁰

1.3

How many electric vehicles are on UK roads?

Cars

The proportion of new electric cars sold in the UK is growing but remains small. Just 3% of cars (1.09 million) were battery electric in the UK at the end of June 2024, and 8% were hybrid electric. 57% were petrol and 32% diesel.⁴¹ Battery electric cars accounted for 19% of all new car registrations at the end of June 2024, an increase from 1% in 2018.⁴²

The growth in ultra-low emission vehicles was initially in plug-in hybrid electric vehicles from 2014 onwards. Growth in battery electric vehicles took off from early 2019 and they outnumbered plug-in electric vehicles from Q4 2020 onwards, as shown in the graph below.

³⁷ Transport Committee, [Fuelling the future: motive power and connectivity](#), 2 March 2023, Report Summary

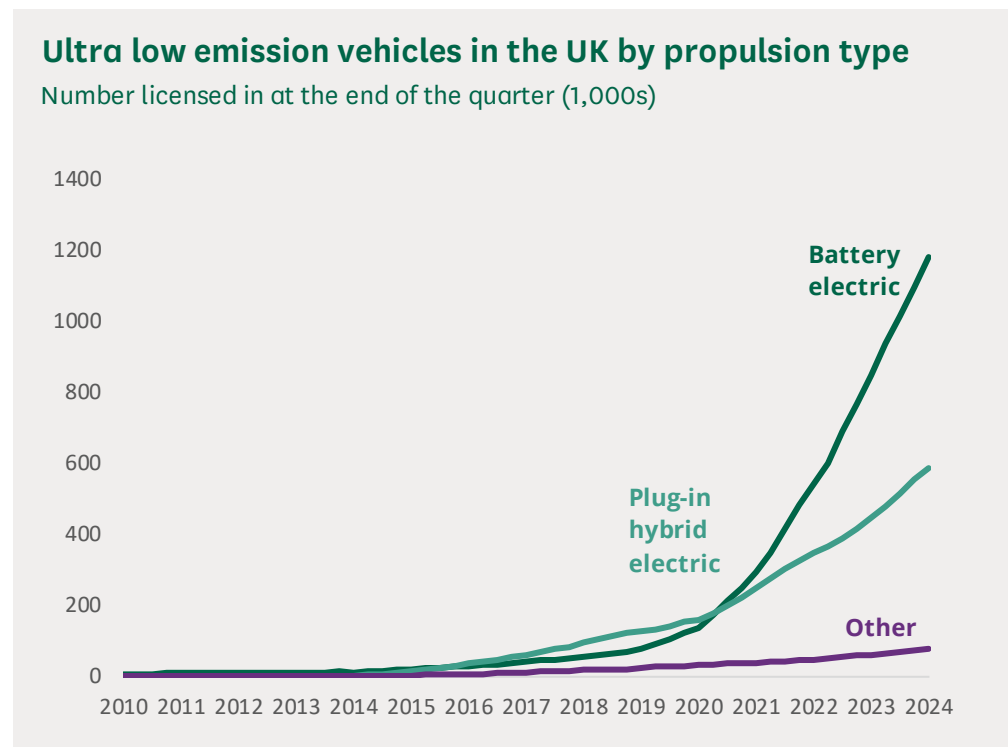
³⁸ Transport Committee, [Fuelling the future: motive power and connectivity](#), 2 March 2023, Report Summary

³⁹ Transport Committee, [Fuelling the future: motive power and connectivity: Government response to the Committee’s Third Report](#), 15 June 2023,

⁴⁰ Office for Zero Emission Vehicles, [Electric vehicles: costs, charging and infrastructure](#), Updated 10 April 2025

⁴¹ DfT, [Vehicle licensing statistics data tables](#), VEH1103

⁴² DfT, [Vehicle licensing statistics data tables](#), VEH0171 and VEH0150



Source: Department for Transport (DfT), [Vehicle licensing statistics](#), VEH0133

Government car fleet

In 2021 the government committed to the entire central government car and van fleet being fully zero emission at the tailpipe by the end of 2027, with a sub-target for 25% of the fleet to be electric by the end of 2022.⁴³ In January 2023 the government confirmed it hit this sub-target, with Ultra Low Emission Vehicles making up 25.5% of its car fleet in September 2022.⁴⁴

Following the 2024 General Election, the government recommitted to 100% of the central government car and van fleet being fully zero emission by 2027.⁴⁵

Taxis

Over the period 2020 to 2024, the proportion of taxis in England using petrol and diesel has fallen (from 89% in 2020 to 71% in 2024). The proportion of hybrid vehicles has more than doubled (from 10% in 2020 to 28% in 2024), and the proportion of battery electric vehicles has increased four-fold (from 0.2% in 2020 to 0.9% in 2024).⁴⁶

⁴³ Department for Environment, Food & Rural Affairs (DEFRA), [Greening government commitments 2021 to 2025](#), 28 October 2021

⁴⁴ DfT, [Government charges ahead in car fleet transition to electric vehicles](#), 19 January 2023

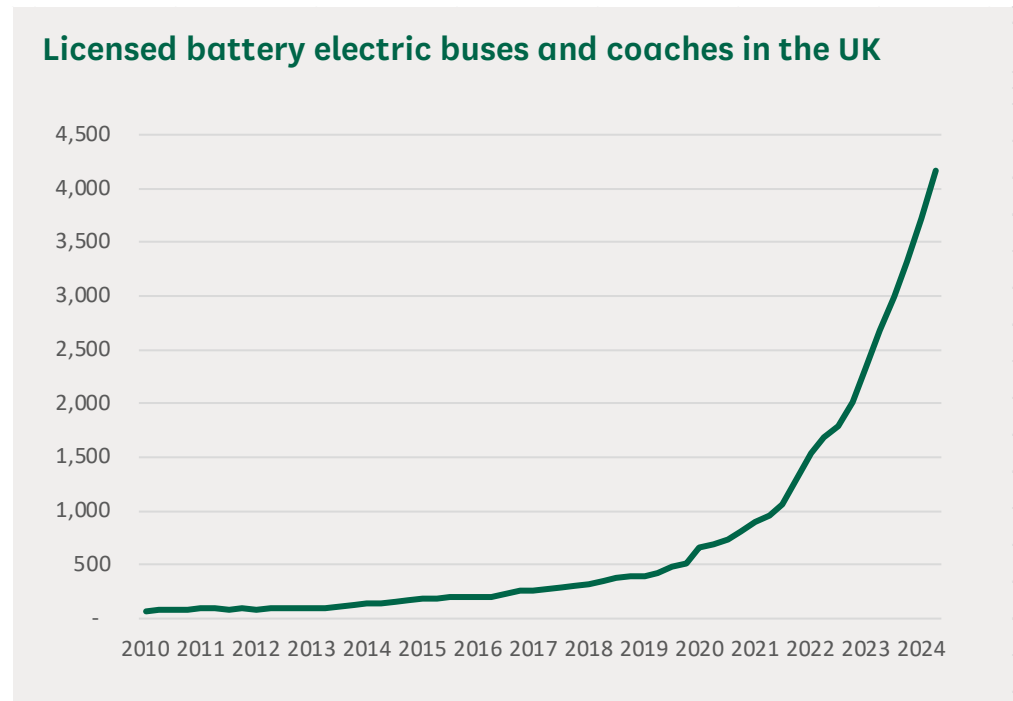
⁴⁵ DfT, [Government fleet commitment](#), updated 17 April 2025

⁴⁶ DfT, [Taxi and private hire vehicle statistics, England: 2024 revised](#), 22 January 2025

⁴⁷ The latest figures are for Q4 2021. The statistics table has been temporarily withdrawn whilst a new methodology is introduced.

Buses

At the end of June 2024, there were 4,160 licensed battery electric buses and coaches in the UK. This equates to 3% of all buses and coaches in the UK.⁴⁸ The chart below shows the number has increased tenfold in the past 6 years from 285 at the end of September 2017.



Source: Department for Transport (DfT), [Vehicle licensing statistics data tables](#), VEH0141a (fuel)

In Great Britain, 8.1% of buses used as Public Service Vehicles by local bus operators were electric (not hybrid) at the end of March 2024.⁴⁹ These figures are not published for the UK.

Lorries and vans

At the end of June 2024, there were 862 battery electric HGVs (heavy goods vehicles) in the UK, which accounted for 0.2% of all HGVs and 73,722 LGVs (light goods vehicles) which accounted for 1.5% of all LGVs.⁵⁰

In November 2024 the SMMT said that, in the third quarter of 2024, battery-electric HGVs accounted for just 0.5% of all new HGV sales.⁵¹

⁴⁸ Department for Transport (DfT), [Vehicle licensing statistics data tables](#), VEH0105 and VEH1103a

⁴⁹ Department for Transport (DfT), [Bus statistics data tables](#), BUS06

⁵⁰ Department for Transport (DfT), [Vehicle licensing statistics data tables](#), VEH0141, VEH0105

⁵¹ SMMT, [UK gets bus boost but green HGVs need backing](#), 15 November 2024

1.4

Electric vehicles in the second-hand market

The UK's used car market is over three times bigger than the new car market. For example, in the first half of 2024 there were 1.0 million new cars compared with 3.9 million used car transactions.⁵²

However, the second-hand market for EVs is less developed. In the first half of 2024, 16.2% of new vehicle registrations were battery electric, 7.8% were plug-in-hybrid and 0.1% were hybrid electric. Whereas only 4.9% of used car transactions were hybrid, 3.3% electric and 1.2% plug-in hybrid in the first quarter of 2025.⁵³

The Society of Motor Manufacturers and Traders (SMMT) reported that battery EV sales increased by over 50% from Q1 2024 to Q1 2025, but remained a small part of the used-car market (1.6% in 2023, up from 0.9% in 2022).⁵⁴ The SMMT called for a VAT cut on new electric vehicles so that supply can grow to meet the demand for second hand electric vehicles.⁵⁵

Over half of all new cars each year are bought by fleet operator businesses. For battery EVs, this figure rises to 83% for the first half of 2024.⁵⁶ These fleet buyers feed through to the used car market. Therefore incentives to such companies to choose EVs when purchasing new vehicles will increase their numbers coming through to the second-hand electric vehicle market.

In its 2024 inquiry into EV strategy, the House of Lords Environment and Climate Change Committee warned that supply is outstripping demand in the used EV market, meaning sellers are cutting prices.⁵⁷ Similarly, car market data analysts Autovista reported that in early 2024 used EVs were losing more of their market value than other cars.⁵⁸ The Lords Committee said that while this trend might be welcomed by second-hand EV buyers in the short-term, it may mean that the cost of leasing a new EV increases, potentially slowing down the EV transition in the longer-term:

We heard that leasing can be helpful for consumers in managing the upfront cost of a vehicle.

Leasing is structured around funding the car's depreciation in value, and monthly leasing repayments are calculated around the expected used price of a vehicle—the greater the depreciation, the higher the amount to fund and

⁵² DfT, [Vehicle Licensing Statistics](#), VE1150; SMMT, [Q1 used car sales surpass two million for first time since 2019](#), 9 May 2025

⁵³ DfT, [Vehicle Licensing Statistics](#), VE0171a and VE0150; SMMT, [Q1 used car sales surpass two million for first time since 2019](#), 9 May 2025

⁵⁴ SMMT, [Used car market up as EV sales reach record levels](#), 8 February 2024; ; SMMT, [Q1 used car sales surpass two million for first time since 2019](#), 9 May 2025

⁵⁵ SMMT, [Used car market up as EV sales reach record levels](#), 8 February 2024

⁵⁶ DfT, [Vehicle Licensing Statistics](#), VE01153

⁵⁷ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, paras 85-86

⁵⁸ Autovista, [Difficult start to 2024 for used BEVs](#), 26 February 2024

therefore the higher the monthly leasing payment for consumers. If used car prices fall, monthly leasing payments will increase.⁵⁹

Some of the reasons for this lack of consumer demand are explored in section 1.6 of this paper.

1.5 How many public chargepoints are there?

In 2022 the government anticipated that by 2030 there will need to be around 300,000 public chargepoints as a minimum in the UK, but there could potentially be double the number.⁶⁰

As of 1 April 2025, there were 76,507 public electric vehicle charging devices installed in the UK, within which:

Charging Speed:

- 15,446 had 50kW of power or above, this represents 20% of all charging devices.⁶¹ These ‘rapid’ chargers can usually fully charge an ‘empty’ EV battery in an hour or less.⁶²
- 42,874 were in the lowest power rating of 3kW up to 8kW, this represents 56% of all charging devices.⁶³ These slower chargers can fully charge an ‘empty’ EV battery between 12 and 22 hours.⁶⁴

Location:

- 37,435 were designated as ‘destination’ chargers, this represents 49% of all charging devices
- 28,316 were designated as ‘on street’ chargers, this represents 37% of all charging devices⁶⁵

Compared to 1 April 2024, 12 months previously:

- there are 16,837 more devices, an increase of 28%⁶⁶

Geographical distribution:

⁵⁹ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, paras 85-86

⁶⁰ DfT, [UK electric vehicle infrastructure strategy](#), 25 March 2022, p2, p38

⁶¹ DfT, [Electric vehicle public charging infrastructure statistics: April 2025](#), 30 April 2025

⁶² Pod Point, [How long does it take to charge an electric car](#), 14 June 2024

⁶³ DfT, [Electric vehicle public charging infrastructure statistics: April 2025](#), 30 April 2025

⁶⁴ Pod Point, [How long does it take to charge an electric car](#), 14 June 2024

⁶⁵ DfT, [Electric vehicle public charging infrastructure statistics: April 2025](#), 30 April 2025

⁶⁶ DfT, [Electric vehicle public charging infrastructure statistics: April 2025](#), 30 April 2025

- There is an uneven geographical distribution of charging devices within the UK. This distribution is more evenly spread for charging devices rated 50kW and above, but some regional disparities remain.
- Some UK local authorities have bid for UK government funding for charging devices, and others have not. Most of the provision of this infrastructure has been market-led, with individual charging networks and other businesses (such as hotels) choosing where to install devices.
- London had a much higher level of charging provision per 100,000 of population than any other part of the UK, with 263 devices per 100,000. In comparison, the average provision in the UK was 113 per 100,000.⁶⁷
- Northern Ireland had the lowest level of charging device provision in the UK, with 35 devices per 100,000, followed by Yorkshire and the Humber with 66 devices per 100,000.⁶⁸

The Commons Library publishes a [dashboard showing the numbers of EVs and public EV chargepoints](#) by local authority.

In addition to the public chargepoint network, it is estimated that there are around 680,000 home chargers installed across the country.⁶⁹ According to a survey conducted by Zap-Map in 2022, about 84% of EV drivers have access to a charging point at home. A vast majority (90%) also use public chargepoints as well, with most utilising them more or less monthly.⁷⁰

Recent government policy on growing the EV chargepoint network can be found in section 2.4 of this briefing.

1.6

Barriers to demand for EVs

Up-front cost

In a 2023 survey from Auto Trader, an online vehicle marketplace, fewer than half of respondents said they would consider switching to an electric car, and that the upfront cost of an EV was the biggest barrier.⁷¹ Auto Trader added that the number of new EV models available for less than £30,000 had decreased from 11 in 2022 to 9 in 2023, whereas there were 87 new diesel and petrol models on sale below £30,000.⁷²

⁶⁷ DfT, [Electric vehicle public charging infrastructure statistics: April 2025](#), 30 April 2025

⁶⁸ DfT, [Electric vehicle public charging infrastructure statistics: April 2025](#), 30 April 2025

⁶⁹ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, para 146

⁷⁰ Installer Online, [Survey: 84% Of EV Drivers Have Access To Domestic Chargers](#), 20 March 2023

⁷¹ “[Fewer than half of drivers willing to make electric vehicle switch](#)”, Auto Trader, 7 September 2023

⁷² “[Fewer than half of drivers willing to make electric vehicle switch](#)”, Auto Trader, 7 September 2023

Auto Trader said EV “uptake is still limited to wealthier drivers” and called for the government to use tax incentives rather than relying on “unsustainable market dynamics”. In particular they called for equal VAT treatment for public and private charging points, which are currently charged at 20% and 5% respectively, making charging more expensive for people without access off-street parking. The government said in answer to a recent PQ that “exceptions to the standard rate [of VAT] have always been limited and balanced against affordability considerations.”⁷³ In October 2024, Sir Christopher Chope introduced a Private Member’s Bill that would exempt public electric vehicle charging points from VAT.⁷⁴

The government said in 2025 that a new EV cost around 40% more than an equivalent ICE vehicle in 2023.⁷⁵ The report said price gap has decreased from 50% in 2020 and predicted that it would fall to 0% in the late 2020s.

The government said in 2022 that EV price was becoming less of a barrier.⁷⁶ In June 2022 it closed the plug-in grant scheme for new electric cars and said it would shift its focus to charging point infrastructure.⁷⁷ After the 2022 Autumn Statement it also legislated to end the vehicle excise duty exemption for EVs from 2025.⁷⁸

In February 2024, in their inquiry into EV strategy, the House of Lords Environment and Climate Change Committee said the removal of incentives to support private buyers with upfront costs for EVs was “premature”.⁷⁹

See section 2.3 of this briefing for more information on financial incentives for EVs.

Charging infrastructure

In 2024 the Lords Environment and Climate Change Committee identified poor access to reliable charging as another key barrier to potential EV buyers. Concerns over charging infrastructure was also identified as a barrier in the 2023 Auto Trader survey.⁸⁰

The Lords committee called for the government to increase the pace of EV chargepoint installation to help the private sector, especially where

⁷³ PQ 45736 [on [Electric Vehicles: Charging Points](#)], 24 April 2025

⁷⁴ [Exemption from Value Added Tax \(Public Electric Vehicle Charging Points\) Bill](#)

⁷⁵ Office for Zero Emission Vehicles, [Electric vehicles: costs, charging and infrastructure](#), Updated 10 April 2025

⁷⁶ PQ 36849 [On [Electric Vehicles: Grants](#)], answered 21 July 2022

⁷⁷ DfT, [Plug-in grant for cars to end as focus moves to improving electric vehicle charging](#), 14 June 2022

⁷⁸ HM Revenue & Customs, [Introduction of Vehicle Excise Duty for zero emission cars, vans and motorcycles from 2025](#), 21 November 2022

⁷⁹ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, summary

⁸⁰ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, summary; “[Fewer than half of drivers willing to make electric vehicle switch](#)”, Auto Trader, 7 September 2023

chargepoints might not be commercially viable. It called for the government to review “outdated and disproportionate planning regulations which are a major block to the rollout”, create legislation to direct local authorities (LAs) who are making insufficient progress on rollout, and remove delays to LA funding for chargepoints.⁸¹

In its response to the inquiry the Conservative government said it was consulting on measures that would give easier street access to chargepoint installers, to speed up rollout.⁸² That consultation ran between February and April 2024.⁸³ In December 2024 the Labour government responded to the consultation and said it would legislate to simplify the process for chargepoint installation.⁸⁴ That legislation is now part of the Planning and Infrastructure Bill, currently going through parliament.⁸⁵

See section 2.4 for more information on charging infrastructure.

Consumer scepticism

The 2024 Lords Environment and Climate Change Committee’s EV inquiry heard that a “concerted campaign of misinformation” about EVs in public discourse was hurting consumer confidence.⁸⁶ The inquiry heard that while some media stories raised legitimate concerns such as those about car manufacturers overstating EVs’ range, others had “alarming headlines” which exaggerated the fire and safety risks of EVs.⁸⁷ Richard Bruce, Director of Transport Decarbonisation at the DfT, told the committee:

I do think there has been an impact from a concerted campaign of misinformation over the last 14 months or so that has been pushing consistent myths about EVs that people absorb and which is reflected in their appetite [for purchasing EVs]. There is an anti-EV story in the papers almost every day. Sometimes there are many stories, almost all of which are based on misconceptions and mistruths, unfortunately.⁸⁸

Some industry commentators have also criticised then Prime Minister Rishi Sunak’s delay to the phase-out date of petrol/diesel cars and vans from 2030 to 2035, announced in September 2023.⁸⁹ Quentin Willson, founder of

⁸¹ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, summary

⁸² Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed: Government response \[PDF\]](#), 19 April 2024

⁸³ DfT consultation, [Street works access: electric vehicle chargepoint operators](#), 5 February 2024

⁸⁴ DfT consultation outcome, [Street works access: electric vehicle chargepoint operators](#), 24 December 2024

⁸⁵ See Commons Library briefing paper CBP-10216, [Planning and Infrastructure Bill 2024-25](#), section 4.4

⁸⁶ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, para 14

⁸⁷ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, para 13

⁸⁸ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, para 14

⁸⁹ Prime Minister’s Office, [PM speech on Net Zero](#), 20 September 2023; Fleet News, [Industry reacts to 2030 new car and van fossil fuel ban delay](#), 21 September 2023

FairCharge, a pro-EV campaign group, told the Transport Committee in May 2024 that the announcement may delay EV purchasing decisions:

I think the mixed messaging from the Government has destabilised the market, absolutely... With the demand taken away, most consumers will say, “Well, I’ll put it off for 11 years to 2035 before I even think about getting an EV.”⁹⁰

The Lords Environment and Climate Change Committee Lords committee recommended government “take a more proactive and leading role in communicating a positive vision of the EV transition to consumers.”⁹¹ In its response, the Government agreed and said it had [published guidance](#) to address common misconceptions about EVs.⁹²

⁹⁰ Transport Committee, [Oral evidence: Electric vehicles](#), HC 742, 15 May 2024, Q47

⁹¹ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, para 29

⁹² Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed: Government response \[PDF\]](#), 19 April 2024, recommendation 1; Office for Zero Emission Vehicles Guidance, [Electric vehicles: costs, charging and infrastructure](#), 5 February 2024

2

Government policy since 2020

The Conservative government under Boris Johnson and Rishi Sunak set phase-out dates for most types of non-zero emission vehicles, provided financial incentives to stimulate demand for zero-emission alternatives, and allocated funding to help the installation of EV chargepoints. It also created specific strategies and funding mechanisms to support the decarbonisation of the bus and freight industries.

The Sunak government legislated for the Zero Emission Vehicle (ZEV) mandate, which gives manufacturers yearly targets for zero-emission car and van production.

The Labour government has largely maintained all these policies although it has made changes to some phase-out dates and modified the ZEV mandate.

2.1

Phase-out dates

In its 2021 Transport Decarbonisation Plan, the government said that it would end the sale of all new non-zero emission road vehicles by 2040 as a ‘backstop’ date.⁹³

For different vehicle types, the government proposed the following ‘phase-out’ dates, which are all earlier than 2040. None of these phase-out dates has been legislated for, although legislation relating to the manufacture of ZEVs has been passed (see section 2.2). Even after the phase-out dates, the use of existing vehicles and the second-hand sales of vehicles would still be permitted. The government under Boris Johnson said that it had “no plans to “ban internal combustion engine vehicles” and the second-hand vehicle market would be unaffected by any phase-out dates.⁹⁴ This position remains unchanged under the Labour government.

The Labour government said in April 2025 that it intended to produce “a plan later in the year for reducing emissions from all sectors, including transport, in line with our legislated carbon budgets”.⁹⁵

⁹³ DfT, [Transport decarbonisation plan](#), 14 July 2021, p5

⁹⁴ [PQ 118031](#), [On Engines], answered 27 November 2020

⁹⁵ [HL Deb 24 April 2025 c754](#)

Cars and vans - 2035

In April 2025 Labour government confirmed an end to the sale of new pure petrol or diesel cars in 2030, which was a manifesto commitment.⁹⁶ It also set an end to new pure petrol-diesel vans in 2035, and that the sale of hybrid cars would be permitted between 2030 and 2035.⁹⁷

This effectively reversed a decision made under the Sunak government and restored the phase-out date introduced under the Johnson government.

In September 2023, then Prime Minister Rishi Sunak announced that 2035 would be the final year that new petrol and diesel cars and vans could be sold in the UK.⁹⁸ This was a change from the previous end date for the sale of new ‘fully’ diesel cars/vans of 2030 which the government under Boris Johnson’s leadership announced on 18 November 2020, following a consultation.⁹⁹

Under the Johnson phase-out plan, hybrid cars could have continued to be sold between 2030 and 2035 “if they have the capability to drive a significant distance with zero emissions.”¹⁰⁰ In contrast, under the Sunak phase-out plan, any petrol/diesel cars or vans, and not just hybrids, could have been sold until 2035.

Under all these plans, car manufacturers would still be expected to increase production of EVs to 80% of total sales by 2030 under the Zero Emission Vehicle (ZEV) Mandate. See section 2.2 of this paper for more on the ZEV Mandate.

Heavy Goods Vehicles – 2035/2040

On 10 November 2021 the government pledged to end the sale of new, non-zero emission HGVs weighing 26 tonnes or less from 2035, and for all new HGVs to be fully zero emission at the tailpipe from 2040.¹⁰¹ This followed a consultation that ran earlier in 2021.¹⁰²

In 2022, the government carried out a call for evidence on potential exemptions to the 2035 phase-out date.¹⁰³ Exemptions could apply to vehicles

⁹⁶ Labour Party, [Change - Labour Party Manifesto 2024](#), p33

⁹⁷ DfT Consultation outcome, [Phasing out sales of new petrol and diesel cars from 2030 and supporting the ZEV transition: summary of responses and joint government response](#), 7 April 2025

⁹⁸ Prime Minister’s Office, [PM speech on Net Zero](#), 20 September 2023

⁹⁹ BEIS, DESNZ, Prime Minister’s Office, [Ten Point Plan for a Green Industrial Revolution](#), 18 November 2020; DfT Consultation, [Consulting on ending the sale of new petrol, diesel and hybrid cars and vans](#), 20 February 2020

¹⁰⁰ DfT, [Government takes historic step towards net-zero with end of sale of new petrol and diesel cars by 2030](#), 18 Nov 2020

¹⁰¹ DfT Press release, [UK confirms pledge for zero-emission HGVs by 2040 and unveils new chargepoint design](#), 10 November 2021

¹⁰² DfT Consultation, [Heavy goods vehicles: ending the sale of new non-zero emission models](#), updated 12 May 2022

¹⁰³ DfT Call for evidence outcome, [Exemptions to 2035 phase-out date for the sale of new non-zero emission HGVs 26 tonnes and under](#), 12 May 2022

operating in remote locations, those carrying abnormal loads, and/or long-haul vehicles. The Conservative government did not respond to this consultation by the end of the last Parliament. The Labour government said in May 2025 that it was “currently considering regulatory approach for decarbonising new non-zero emission heavy goods vehicles”.¹⁰⁴

Motorcycles and ‘L-category vehicles’ – 2035

L-category vehicles include mopeds and motorcycles, motor tricycles, quad bikes and quadricycles.¹⁰⁵

In a 2022 consultation, the government proposed ending new non-zero emission L-category vehicles sales by 2035, and by 2030 for the most lightweight of these vehicle types.¹⁰⁶ The Conservative government did not respond to this consultation by the end of the last Parliament. In February 2025 the Labour government said it would “set out any plans to decarbonise motorcycles in due course”.¹⁰⁷

Buses and coaches

The Conservative government did not set a date for the end of sales of new non-zero emission buses or coaches. It carried out a consultation on the topic in 2022 which called for views “on setting a specific date between 2025-2032 for ending the sale of new non-zero emission (at the tailpipe) buses.”¹⁰⁸ The Conservative government did not respond to this consultation by the end of the last Parliament.

The Labour government’s Bus Services (No. 2) Bill, currently going through parliament, would require bus operators in England to stop registering new non-zero-emission buses after a specified date to be confirmed by regulations. This date cannot be earlier than 1 January 2030.¹⁰⁹

2.2

Zero Emission Vehicle (ZEV) mandate

The Zero Emission Vehicle (ZEV) mandate is intended to stimulate UK vehicle manufacturers to sell more electric vehicles (EVs) year on year. It came into force in January 2024 under the Conservative government.¹¹⁰ When in

¹⁰⁴ PQ 52652 [[Large Goods Vehicles: Carbon Emissions](#)], answered 21 May 2025

¹⁰⁵ DfT Closed consultation, [L-category vehicles: ending sales of new non-zero emission models](#), 14 July 2022, para 1.12

¹⁰⁶ DfT Closed consultation, [L-category vehicles: ending sales of new non-zero emission models](#), 14 July 2022

¹⁰⁷ PQ HL4280 [[On Motorcycles: Petrol](#)], answered 4 February 2025

¹⁰⁸ DfT Closed consultation, [Ending the sale of new, non-zero emission buses, coaches and minibuses](#), 26 March 2022

¹⁰⁹ Commons Library research briefing CBP-10266, [Bus Services \(No. 2\) Bill \[HL\] 2024–25](#)

¹¹⁰ DfT Press Release, [Pathway for zero emission vehicle transition by 2035 becomes law](#), 3 January 2024

opposition, the Labour Party supported the ZEV mandate in Parliament.¹¹¹ In government, Labour has continued it, describing it in November 2024 as “the largest carbon saving measure across government.”¹¹²

Policy development

On 14 July 2021, alongside the publication of its transport decarbonisation plan, the government published a consultation on a regulatory framework for vehicle manufacturers to stimulate more vehicle decarbonisation and “address barriers, including certainty of vehicle supply.”¹¹³

In April 2022 the government responded to the consultation, confirming it would proceed with a Zero Emission Vehicle (ZEV) Mandate scheme for cars and vans, something that the Climate Change Committee had previously recommended.¹¹⁴ The government said that such a scheme could potentially be used for other vehicle types as well in future.¹¹⁵ At the same time it launched a technical consultation on how such a scheme might operate.¹¹⁶

In March 2023 the government responded to the technical consultation, and launched a consultation on the final version of a UK-wide ZEV Mandate scheme.¹¹⁷ The ZEV mandate relates to climate policy and is therefore a devolved matter, so the consultation was made in co-ordination with the Welsh Government, Scottish Government and the Northern Ireland Department for Infrastructure (DfI).¹¹⁸

In September 2023 the government and devolved administrations responded to the consultation and announced the final confirmed ZEV Mandate which would be launched from 2024.¹¹⁹

In 2023, the Labour Party voted for ZEV mandate when it was legislated for in Parliament.¹²⁰

¹¹¹ [HC Deb Monday 4 December 2023](#), Division 25.

¹¹² PQ 11372 [[Motor Vehicles: Carbon Emissions](#)] 5 November 2024

¹¹³ DfT Closed Consultation, [CO2 emissions regulatory framework for all newly sold road vehicles in the UK](#), 14 July 2021; DfT, [Transport decarbonisation plan](#), 14 July 2021

¹¹⁴ CCC, [The UK’s transition to electric vehicles](#), 9 December 2020

¹¹⁵ DfT Closed Consultation, [CO2 emissions regulatory framework for all newly sold road vehicles in the UK](#), 7 April 2022

¹¹⁶ DfT Consultation outcome, [Policy design features for the car and van zero emission vehicle \(ZEV\) mandate](#), 30 March 2023

¹¹⁷ DfT Consultation outcome, [Policy design features for the car and van zero emission vehicle \(ZEV\) mandate](#), 30 March 2023; DfT Consultation outcome, [A zero emission vehicle \(ZEV\) mandate and CO2 emissions regulation for new cars and vans in the UK](#), 28 September 2023

¹¹⁸ [The Vehicle Emissions Trading Schemes Order 2023](#), Explanatory Notes

¹¹⁹ DfT Consultation outcome, [A zero emission vehicle \(ZEV\) mandate and CO2 emissions regulation for new cars and vans in the UK](#), 28 September 2023

¹²⁰ [HC Deb Monday 4 December 2023](#), Division 25. [The Vehicle Emissions Trading Schemes Order 2023](#)

What is the ZEV mandate?

The [Zero Emission Vehicle \(ZEV\) mandate](#) for car manufacturers came into force in January 2024.¹²¹

The mandate phases out the sale of new non-zero emission cars and vans. It specifies the minimum percentage of car manufacturers' sales that must be zero-emission vehicles each year, rising from 22% in 2024, to 80% by 2030 and 100% in 2035. The table below shows the yearly targets for zero-emission car and van sales each year to 2035.

The ZEV mandate was legislated for via [The Vehicle Emissions Trading Schemes Order 2023](#), which sets targets up to 2030.¹²² The government and devolved administrations said that targets for the period 2030 to 2035 will be legislated for "later in the decade".¹²³ The legislation currently applies only in Great Britain. On 22 May 2024 the Northern Ireland Assembly indicated that it would also implement the mandate from January 2025.¹²⁴

ZEV mandate target dates						
	2024	2025	2026	2027	2028	2029
Cars	22%	28%	33%	38%	52%	66%
Vans	10%	16%	24%	34%	46%	58%
	2030	2031	2032	2033	2034	2035
Cars	80%	84%	88%	92%	96%	100%
Vans	70%	76%	82%	88%	94%	100%

Notes: targets for 2031-2035 are yet to be set in legislation.

Source: [The Vehicle Emissions Trading Schemes Order 2023](#) Schedule 6; DfT, [Zero emission vehicle \(ZEV\) mandate consultation: summary of responses and joint government response](#)

The legislation also includes a CO₂ emissions standard for non-ZEV cars and vans. This is to address a recent trend where emissions reductions gained from increasing ZEV sales are offset by CO₂ emissions increasing across the rest of the fleet.¹²⁵ Manufacturers receive a baseline target for the CO₂

¹²¹ DfT Press Release, [Pathway for zero emission vehicle transition by 2035 becomes law](#), 3 January 2024

¹²² [The Vehicle Emissions Trading Schemes Order 2023](#) was passed using powers in [Part 3 of the Climate Change Act 2008](#)

¹²³ DfT, [Zero emission vehicle \(ZEV\) mandate consultation: summary of responses and joint government response](#), 25 October 2023

¹²⁴ NI Assembly Committee for Infrastructure, [Minutes of Proceedings](#), 22 May 2024

¹²⁵ DfT, [Decarbonising transport: Greener, Better Britain](#), July 2021, p14

emissions allowed for the non-ZEV proportion of their fleet based on 2021 data; this target is fixed at the same level to 2030.¹²⁶

Together, the ZEV mandate and the CO₂ emissions standard are called the Vehicle Emissions Trading Schemes (VETS).

How does the ZEV mandate work?

The ZEV mandate and CO₂ emissions standard are trading schemes. For each non-ZEV sold, the manufacturer must have an allowance.

Manufacturers that sell more ZEVs than the regulations require will have a surplus of allowances that they can sell, bank (for use in later years) or convert (to offset against the CO₂ emissions for the rest of the fleet).

If a manufacturer wants to sell more non-ZEVs than the target, they will need to either get more allowances (by trading) or pay a 'compliance payment'. Manufacturers can get more allowances by buying them from another manufacturer or borrowing them from a future year. They can also convert overperformance on the CO₂ emissions allowance into ZEV-allowances.¹²⁷

There are caps and restrictions on some of the trading mechanisms. For example, banked allowances have a 3-year time limit.¹²⁸ There are exemptions for small vehicle manufacturers and special purpose vehicles (for example, wheelchair accessible vehicles).¹²⁹

The compliance payments are:

- £15,000 per non-ZEV car sold without an allowance.
- £9,000 per non-ZEV van without an allowance in 2024, rising to £18,000 for 2025 onwards.¹³⁰

¹²⁶ CO₂ emissions standards have been in place for UK car manufacturers since 2009. Prior to Brexit these standards were set at EU level. The EU has adopted a system of gradually decreasing CO₂ emissions standards. The EU regulations in place at the end of the Brexit transition period (31 December 2020) were retained in UK law. A [DfT consultation on CO₂ emission standards for passenger cars](#) (October 2020) provides further information. [The Vehicle Emissions Trading Schemes Order 2023](#) amends the retained CO₂ emissions standards, replacing it with the ZEV mandate and a fixed CO₂ standard to 2030. The CO₂ standard is fixed rather than decreasing, to encourage manufacturers to focus investments on ZEV vehicles.

¹²⁷ DfT, [Zero emission vehicle \(ZEV\) mandate consultation: summary of responses and joint government response](#), 25 October 2023, Overview of responses.

¹²⁸ DfT, [Zero emission vehicle \(ZEV\) mandate consultation: summary of responses and joint government response](#), 25 October 2023, question 8-9.

¹²⁹ DfT, [Zero emission vehicle \(ZEV\) mandate consultation: summary of responses and joint government response](#), 25 October 2023, questions 5-6.

¹³⁰ DfT, [Zero emission vehicle \(ZEV\) mandate consultation: summary of responses and joint government response](#), 25 October 2023, question 13.

In April 2025 the government said it will reduce these compliance payment rates.¹³¹ See the section on 2025 reforms to the ZEV Mandate³ below.

Compliance payments for the CO₂ emissions standard are £86 per gram of CO₂ above the manufacturers target, multiplied by the number of non-ZEV cars or vans sold.¹³²

Criteria for zero-emissions vehicles

A zero-emission vehicle is defined as any vehicle with zero CO₂ emissions from the tailpipe – any propulsion technology meeting this standard can qualify (see section 1.2). Other greenhouse gases (such as NO_x) are not included in the requirements, as there are no internationally agreed testing regimes for cars that test and certify for these.¹³³

Zero-emission cars and vans must have a minimum electric range of 100 miles to qualify as a compliant sale. Manufacturers must also provide the following minimum warranties on ZEV cars and vans:

- a warranty period of at least 8 years or 100,000 miles (whichever reached first) on the battery (or hydrogen fuel cell stack or hydrogen tank)
- provide a replacement battery if it falls below 70% capacity during the warranty period (65% for vans)
- a warranty period of at least three years or the first 60,000 miles (whichever reached first) for the rest of the vehicle.¹³⁴

Commentary on the ZEV mandate

The Conservative government said in 2023 that the UK ZEV mandate is the “most ambitious regulation of its kind” in the world.¹³⁵ The Labour government described it in November 2024 as “the largest carbon saving measure across government.”¹³⁶

A similar mandate scheme operates in California (and some other US States and Canadian provinces), however these schemes allow plug-in hybrids to

¹³¹ DfT Consultation outcome, [Phasing out sales of new petrol and diesel cars from 2030 and supporting the ZEV transition: summary of responses and joint government response](#), 7 April 2025

¹³² DfT, [Zero emission vehicle \(ZEV\) mandate consultation: summary of responses and joint government response](#), 25 October 2023, question 22.

¹³³ Zero grams CO₂/ km according to the [Worldwide Harmonized Light Vehicles Test Procedure](#) (WLTP); DfT, [Zero emission vehicle \(ZEV\) mandate consultation: summary of responses and joint government response](#), 25 October 2023, question 9.

¹³⁴ [The Vehicle Emissions Trading Schemes Order 2023 SI 2023/1394, article 3, paragraph 7-9.](#)

¹³⁵ DfT, [Zero emission vehicle \(ZEV\) mandate consultation: summary of responses and joint government response](#), 25 October 2023

¹³⁶ PQ 11372 [[Motor Vehicles: Carbon Emissions](#)] 5 November 2024

count towards compliance.¹³⁷ The EU has chosen a system of CO₂ emissions regulation, where car manufacturers are gradually required to reduce their fleet-wide emissions, reaching zero emissions in 2035.¹³⁸

The government said in 2023 that the ZEV mandate provides “certainty to manufacturers, chargepoint operators, and consumers” and sets a “clear pathway” to end the sale of non-ZEV cars and vans in 2035”.¹³⁹

The purpose and aims of the ZEV mandate have been broadly welcomed. Before the government announced it would proceed with a ZEV mandate, the Climate Change Committee, Transport & Environment (a campaign group for clean transport), Policy Exchange and the Green Alliance had all suggested such a mandate was necessary to meet the UK’s net zero targets.¹⁴⁰

The Society of Motor Manufacturers and Traders (SMMT), trade body for the car industry, said the industry recognised the importance of the mandate and was “investing billions in decarbonisation”.¹⁴¹

2025 reforms to ZEV Mandate

In November 2024, Stellantis announced the closure of its car factory in Luton, citing the ZEV mandate as a contributing factor to its decision.¹⁴²

On 27 November, the Business Secretary Jonathan Reynolds told the Commons that the government would launch a “fast-track” consultation on implementing Labour’s manifesto commitment to phase out pure petrol/diesel cars by 2030.¹⁴³ The consultation would also propose adding flexibilities to the ZEV mandate to reassure car manufacturers:

We will shortly fast-track a consultation on our manifesto commitment to ending the sales of new pure petrol and diesel cars by 2030.

We will use that consultation to engage with industry on the previous Government’s ZEV transition mandate, and the flexibilities in it, and we will welcome the industry’s feedback as we move forwards.¹⁴⁴

¹³⁷ California Air Resources Board, [Zero Emissions Vehicles Programme: About](#) [accessed 26 June 2024].

¹³⁸ European Parliament, [Deal confirms zero-emissions target for new cars and vans in 2035](#), 27 October 2022; [Regulation \(EU\) 2023/851](#) 19 April 2023.

¹³⁹ DfT, [Zero emission vehicle \(ZEV\) mandate consultation: summary of responses and joint government response](#), 25 October 2023

¹⁴⁰ The Climate Change Committee, [The UK’s transition to electric vehicles](#) [PDF], December 2020; Policy Exchange, Route ‘35 [How a California-style ZEV Mandate can deliver the phase out of petrol and diesel cars](#) [PDF], July 2020; Transport & Environment, [Phasing out sales of new cars with engines](#) [PDF], July 2020; Green Alliance, [Accelerating the electric vehicle revolution: Why the UK needs a ZEV mandate](#) [PDF], May 2021.

¹⁴¹ SMMT, [SMMT statement on the Zero Emission Vehicle Mandate](#), 28 September 2023.

¹⁴² BBC News, [“Vauxhall owner to close Luton factory”](#), 26 November 2024

¹⁴³ Labour Party, [Change - Labour Party Manifesto 2024](#), p33

¹⁴⁴ [HC Deb 27 November 2024 c7776](#)

The consultation ran between December 2024 and February 2025. Responding to the consultation in April 2025, the government said that the car industry had exceeded its ZEV mandate targets for 2024, but the van industry had not, and said there was a need to respond to industry pressures:

In 2024, the first year of the Zero Emission Vehicle (ZEV) Mandate, vehicle manufacturers rose to the challenge, achieving a compliance rate of 24.3% for the year – exceeding the 22% target. We recognise the van market faced some challenges and fell slightly short of the ZEV Mandate trajectory, reaching a compliance rate of 9.3% – just below the 10% target.

The government recognises the need to be bold and ambitious to meet our climate change goals, while equally understanding the wider pressures that are being faced by industry. As such, this response contains policy decisions that carefully consider these challenges.¹⁴⁵

The response confirmed an end to the sale of new pure petrol-diesel cars in 2030, an end to new pure petrol-diesel vans in 2035, and that the sale of hybrid cars would be permitted between 2030 and 2035. On the ZEV mandate, the following flexibilities were confirmed:

- The ability for manufacturers to borrow ZEV allowances from future years would be extended from 2027 to 2029.
- The ability to transfer CO₂ savings from non-ZEV to ZEVs would be extended to 2029.
- The introduction of a new bidirectional transfer mechanism to allow ‘excess’ ZEV cars and van sales to be transferred between car/van quotas, with an exchange rate for ZEV car to ZEV van of 0.4; for ZEV van to ZEV car of 2.
- A reduction in compliance payments from £15,000 to £12,000 for every ZEV short of a car manufacturer’s target.¹⁴⁶

It is not yet clear when the ZEV mandate and the wider VETS will be amended in law.

Reaction to the ZEV Mandate reforms

Writing in the Times, the Prime Minister Sir Keir Starmer said the previous government had designed the ZEV mandate “with nowhere near enough regard for manufacturing” and the reforms meant the Labour government was giving manufacturers “the flexibility they need to hit the target[s] in a way that works for them while keeping our manifesto commitment to end sales of petrol cars by 2030.”¹⁴⁷

¹⁴⁵ DfT Consultation outcome, [Phasing out sales of new petrol and diesel cars from 2030 and supporting the ZEV transition: summary of responses and joint government response](#), 7 April 2025

¹⁴⁶ DfT Consultation outcome, [Phasing out sales of new petrol and diesel cars from 2030 and supporting the ZEV transition: summary of responses and joint government response](#), 7 April 2025

¹⁴⁷ Sir Keir Starmer, “[My only priority is making British people better off](#)”, The Times, 6 April 2025

The industry body, the Society of Motor Manufacturers and Traders (SMMT) welcomed the reforms to the ZEV mandate, saying the government “had rightly listened to industry”, but it needed to do more to support customers of electric cars and vans through fiscal incentives.¹⁴⁸

Environmental campaign group Transport & Environment said the reforms would be “bad news for consumers as it will delay price reductions and the supply of affordable EVs”.¹⁴⁹ The NGO Green Alliance said that the new flexibilities in the ZEV mandate would create more uncertainty about the pace of the transition and could reduce investor confidence.¹⁵⁰

¹⁴⁸ SMMT, [Government announces adjustments to ZEV Mandate](#), 10 April 2025

¹⁴⁹ Transport & Environment, [Weakening the ZEV Mandate is counterproductive](#), 7 April 2025

¹⁵⁰ Green Alliance, [Don't put the brakes on the UK's market lead in electric vehicles](#), 29 April 2025

2.3

Financial incentives for low-emission vehicles

Grants

EVs are currently more expensive than equivalent petrol or diesel engine vehicles and are not projected to reach price parity until the late-2020s.¹⁵¹ To help bridge the gap, the government provides “plug-in grants” which reduce the cost of new EVs, as described in the table below.

Funding for these grants was due to expire in April 2025, but in the October 2024 budget the government extended funding until April 2026 and made some minor changes to grants.¹⁵²

Table 1 Low-emission vehicles eligible for a plug-in grant

Category	CO2 emissions	Zero emission range	Grant	Maximum amount
Motorcycles (up to £10,000)	No CO ₂	50 km (31 miles)	35%	Up to £500
Small vans (up to 2,500 kg)	Less than 50 g/km	96 km (60 miles)	35%	Up to £2,500
Large vans (2,500 kg – 4,250 kg)	Less than 50 g/km	96 km (60 miles)	35%	Up to £5,000
Small trucks (4,250kg – 12,000 kg)	Less than 50% of equivalent Euro VI	96 km (60 miles)	20%	Up to £16,000
Large trucks (more than 12,000 kg)	Less than 50% of equivalent Euro VI	96 km (60 miles)	20%	Up to £25,000
Taxis	Less than 50 g/km	112 km (70 miles)	20%	Up to £4,000
Wheelchair accessible vehicles (up to £50,000)	No CO ₂	112 km (70 miles)	35%	Up to £2,500

Source: DfT, [Low-emission vehicles eligible for a plug-in grant](#). (Accessed 21 May 2025)

Vehicle buyers do not apply for the grant. The seller includes it as a discount in the purchase price.¹⁵³

On 14 June 2022, the government closed the plug-in grant for private cars to new orders, saying that successive reductions in the size of the grant, and the number of models it covers, have had little effect on sales or the range of

¹⁵¹ Transport & Environment, [EVs will be cheaper than petrol cars in all segments by 2027](#), May 10 2021

¹⁵² HM Treasury, [Autumn Budget 2024](#), October 2024; DfT Press release, [£120 million to roll-out more electric vans, taxis and motorbikes](#), 25 February 2025

¹⁵³ [Gov.uk: Low-emission vehicles eligible for a plug-in grant](#). (accessed 21 May 2025)

models being manufactured.¹⁵⁴ The government said that funding would be refocussed on public charging and supporting the purchase of other road vehicles, such as taxis, vans and motorcycles.¹⁵⁵

In its 2024 inquiry into EV strategy, the House of Lords Environment and Climate Change Committee called the removal of incentives “premature”:

The upfront cost of EVs, including second-hand cars, remains a significant barrier to consumer adoption and targeted grants should be reconsidered for EV purchases. Unlike markets across Europe the Government has removed incentives to support private buyers with upfront costs for EVs. This is premature as we seek to move from early adopters and fleets purchasing EVs to a wider proportion of the population. Any incentives should be tapered as the prices of EVs fall and approach parity with petrol and diesel equivalents.¹⁵⁶

Scottish Government initiatives

Transport Scotland, part of the Scottish Government, previously provided funding for two schemes which have both now closed. The first was an electric vehicle interest free loan up to £28,000 for purchasing new electric vehicles. The second was a used electric vehicle loan, providing £20,000 or £5,000 for used electric cars or motorcycles/mopeds respectively.¹⁵⁷

Fiscal incentives

From 1 April 2025, vehicle excise duty (VED) applies to zero-emission vehicles. The expensive car supplement (a VED surcharge applicable to new cars with a list price exceeding £40,000) also began applying to EVs from that date.¹⁵⁸

Until 31 March 2025, EVs were exempt from VED and from paying the expensive car supplement.¹⁵⁹

In 2024 the Society of Motor Manufacturers and Traders (SMMT) told the House of Lords Environment and Climate Change Committee that it was concerned about the removal of this exemption. It said the removal will capture as much as half the BEV market, and add around £3,000 to the cost of the vehicle over the first six years of ownership, “notably hitting years 4-6,

¹⁵⁴ DfT, [Plug-in grant for cars to end as focus moves to improving electric vehicle charging](#), 14 June 2022

¹⁵⁵ DfT, [Plug-in grant for cars to end as focus moves to improving electric vehicle charging](#), 14 June 2022

¹⁵⁶ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, Summary, p5

¹⁵⁷ Energy Saving Trust, [Electric Vehicle Loan](#), accessed on 28 April 2021; Energy Saving Trust, [Used Electric Vehicle Loan](#), accessed on 28 April 2021

¹⁵⁸ DVLA, [Vehicle tax rates \(V149 and V149/1\)](#), updated 1 April 2025

¹⁵⁹ HM Revenue & Customs, [Introduction of Vehicle Excise Duty for zero emission cars, vans and motorcycles from 2025](#), 21 November 2022

when it would typically be in the used car market, which could limit demand and so residual values”.¹⁶⁰

Currently EVs are charged VAT at the 20% standard rate. The SMMT also called for a VAT reduction for EVs in line with other “environmental products” such as solar panels and heat pumps which can be zero-rated for VAT in certain circumstances.¹⁶¹

The box below explains how VED and the expensive car supplement have been used to incentivise the purchase of EVs in recent years.

Fiscal incentives and VED

VED is an annual tax levied for most types of vehicles to be used (or kept) on public roads. Certain vehicles are exempt from paying VED.

For cars first registered between 1 March 2001 and 31 March 2017, VED rates were linked to emissions, meaning lower emission cars pay lower rates. Cars that emitted less than 100 g/km of carbon dioxide were exempt from VED. Rates for other vehicles were on a sliding scale, with the most polluting paying the highest levels of tax.¹⁶²

Major reforms to VED were introduced in the 2015 Budget (taking effect from 1 April 2017). From that date, only cars that emit 0 g/km of carbon dioxide continued to be exempt. All other vehicles began having to pay a first-year rate on the year they are first registered, which is still linked to carbon emissions. For instance, in 2025/26, cars emitting between 1 and 50 gCO₂/km have to pay a first-year rate of £10 – the amount increases the higher the carbon emissions rate gets. From the second year onwards, all cars that emit more than 0 gCO₂/km pay a flat rate of £195/year.¹⁶³

All cars with a list price greater than £40,000 also pay a supplement of £425/year, from the second to the sixth year since

¹⁶⁰ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, [ELV0117](#), para 16

¹⁶¹ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, [ELV0117](#), para 15

¹⁶² DVLA, [Vehicle tax rates \(V149 and V149/1\)](#), updated 1 April 2025

¹⁶³ DVLA, [Vehicle tax rates \(V149 and V149/1\)](#), updated 1 April 2025. Note that VED rates are usually uprated with inflation each year, see HM Revenue & Customs, [Vehicle Excise Duty rates for cars, vans and motorcycles from April 2024](#), 22 November 2023

purchase.¹⁶⁴ At Budget 2020, zero emission vehicles were exempted from paying this “expensive car supplement”.¹⁶⁵

In practice the strict emissions limit means that only fully electric vehicles qualified for the exemption, and the flat rate for other vehicles has removed the incentive to purchase alternative ultra-low emissions vehicles, such as plug-in hybrid EVs or hydrogen fuel cell cars. Further, as hybrid vehicles are more expensive than equivalent conventional models, they are disproportionately affected by the VED supplement.

In the 2022 Autumn Statement, the Conservative government announced that EVs would no longer be exempt from VED from 1 April 2025. Instead, they will be liable to pay the lowest rate in the first year and the standard rate from the second year. EVs would also lose their exemption from paying the expensive car supplement in 2025 for vehicles over £40,000 in years two to six. Zero emission vans and motorbikes would also move to the petrol/diesel rate for light vehicles and smallest engine size rate respectively.¹⁶⁶ The Conservative government introduced legislation to this effect. These provisions are now included in [section 10 of the Finance Act 2023](#).

At Autumn Budget 2024, Chancellor Rachel Reeves said that the gap between the first-year VED rate for EVs and for other cars would increase, to maintain an incentive to purchase zero-emission cars.¹⁶⁷ The Budget document also said the government would consider increasing the £40,000 threshold for the expensive car supplement for EVs, with a decision to be made at a future fiscal event.¹⁶⁸

The Library briefing [Vehicle excise duty \(VED\)](#) provides an extensive background and overview of the evolution of VED. The Library briefing [Vehicle excise duty and zero emission vehicles](#) has further detail of the decision to remove the VED exemption on EVs.

Salary sacrifice and company car tax

Electric car salary sacrifice schemes are offered by some employers. Under such schemes, a portion of an employee’s earnings before tax is automatically deducted and used to fund leasing repayments on an EV. This

¹⁶⁴ DVLA, [Vehicle tax rates \(V149 and V149/1\)](#), updated 1 April 2025

¹⁶⁵ HM Treasury, [Budget 2020](#) [PDF], Mar 2020, para 1.245.

¹⁶⁶ HM Treasury, [Autumn Statement 2022](#), 17 November 2022

¹⁶⁷ HC Deb [30 October 2024](#), c819

¹⁶⁸ HM Treasury, [Autumn Budget 2024](#), October 2024

reduces the amount of income tax and national insurance paid. The EV is leased from a third-party firm by the employer, who in turn leases it to the employee under the scheme. Servicing and insurance are typically included as part of the scheme.¹⁶⁹

Company car tax is paid by an employee who drives a company car privately (including for commuting).¹⁷⁰ Generally, individuals are taxed on the cash value of benefits they enjoy because of their employment. The cash value of these benefits is added onto their taxable income and taxed accordingly. For company cars, the cash value is calculated as a percentage of a car's price, based on CO₂ emissions.

Drivers of battery EVs pay significantly lower benefit in kind rates at 2 per cent, compared to up to 37 per cent for the most polluting cars.¹⁷¹ At Autumn Budget 2024, the government said company car tax for EVs will increase 1% each year to a total of 5% in April 2028, and then 2% each year to a total of 9% in 2029/30. By contrast, the most polluting cars (emitting over 160 grams of CO₂ per kilometre) will pay 39% company car tax in 2029/30.¹⁷²

In its 2024 inquiry into EV strategy, the House of Lords Environment and Climate Change Committee heard that this lower benefit in kind rate has been “the single most effective intervention to date” in incentivising the uptake of EVs.¹⁷³ This should be seen in the context that just over half of all new cars each year are bought by fleet operator businesses whereas for battery EVs, this figure rises to 81 per cent.¹⁷⁴ Therefore incentives to such companies to choose EVs when purchasing new vehicles can impact the whole second-hand EV market.

2.4 Improving charging infrastructure

The government's March 2022 Electric Vehicle Infrastructure Strategy identified charging infrastructure as “the single biggest challenge” of the EV transition.¹⁷⁵ The strategy set out a vision to:

¹⁶⁹ HM Treasury, [EIM42750 - Salary sacrifice: what is a salary sacrifice: arrangement of guidance](#), updated 14 May 2025; Carwow, ‘[Electric car salary sacrifice scheme explained](#)’, 26 January 2023; Octopus Electric Vehicles, ‘[How does Benefit in Kind affect electric cars?](#)’, 12 December 2023

¹⁷⁰ HM Government, [Tax on company benefits](#), accessed 29 May 2025

¹⁷¹ Office for Zero Emission Vehicles Guidance, [Electric vehicles: costs, charging and infrastructure](#), 5 February 2024; Tax is paid if the company car is used privately (including for commuting), see Gov.uk, [Tax on company benefits: Tax on company cars](#) (accessed 15 May 2025) and HMRC, [Autumn Budget 2024: Overview of tax legislation and rates – Annex A: rates and allowances](#), updated 11 November 2024

¹⁷² HMRC, [Autumn Budget 2024: Overview of tax legislation and rates – Annex A: rates and allowances](#), updated 11 November 2024

¹⁷³ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, para 90

¹⁷⁴ DfT, [Vehicle licensing statistics: 2023](#), 15 May 2024

¹⁷⁵ DfT, [UK electric vehicle infrastructure strategy](#), 25 March 2022, p9

remove charging infrastructure as both a perceived, and a real, barrier to the adoption of electric vehicles (EVs). EV charging should be cheaper and more convenient than refuelling at a petrol station.¹⁷⁶

In the strategy the government said that most charging will take place at home, but that public charging is needed both to facilitate longer journeys and for people without off-street parking.¹⁷⁷

In 2022 the government said that by 2030 there will need to be around 300,000 public chargepoints as a minimum in the UK, but that there could potentially be double the number.¹⁷⁸ In December 2024 the National Audit Office said the government was broadly on track to meet its EV chargepoint targets, but that there were some important geographical differences and problems with rapid charger provision.¹⁷⁹

Grants for private chargepoints

To ensure EV owners can enjoy one of the “key attractions” of owning an EV¹⁸⁰ – home charging – the government’s Office for Zero Emission Vehicles (OZEV) created a range of grants to support the installation of private chargepoints.

The following UK-wide grant schemes are available to individuals and certain employers to apply to. All provide up to 75% (capped at £350) towards the cost of installing an EV chargepoint. All the grant funding was due to expire in March 2025 but at the October 2024 budget the government extended this to 31 March 2026.¹⁸¹

The maximum number of charge points permitted and other criteria differ between schemes:

- [chargepoint grant for renters and flat owners](#)
- [chargepoint grant for households with on-street parking](#) (see below)
- [chargepoint and infrastructure grant for business staff and fleet car parks](#)
- [chargepoint and infrastructure grant for landlords, including car parks](#)
- [Workplace Charging Scheme](#)
- [Workplace Charging Scheme for state-funded education institutions](#)¹⁸²

¹⁷⁶ DfT, [UK electric vehicle infrastructure strategy](#), 25 March 2022, p4

¹⁷⁷ DfT, [UK electric vehicle infrastructure strategy](#), 25 March 2022, p4

¹⁷⁸ DfT, [UK electric vehicle infrastructure strategy](#), 25 March 2022, p2, p38

¹⁷⁹ NAO, [Public chargepoints for electric vehicles](#), 13 December 2024

¹⁸⁰ DfT, [Reducing emissions from road transport: Road to Zero Strategy](#), July 2018, p83

¹⁸¹ HM Treasury, [Autumn Budget 2024 \(HTML\)](#), 30 October 2024

¹⁸² Office for Zero Emission Vehicles, [Electric vehicle chargepoint and infrastructure grant guidance for installers](#), updated 18 March 2024

Cross-pavement charging

Pavements, like roads, are the responsibility of the local highway authority, which is generally the unitary or county council. Many highway authorities do not permit EV charging cables being run from a property, across a pavement, to an EV (even if a cable cover or mat is used).¹⁸³ This may constitute a trip hazard and an obstruction under the Highways Act 1980.¹⁸⁴

In March 2024 the government made grants available for drivers who do not have a driveway but do have access to on-street parking. The grant will part-fund an EV chargepoint using a cross-pavement charging solution, such as a recessed cable channel/gully which goes over the top of a pavement, or a cable that goes fully underneath the pavement). However, installing this kind of cross-pavement solution requires the approval of the local highway authority.¹⁸⁵ Many councils will not give such permission, because running a cable across the pavement can present a safety hazard for pedestrians or act as a barrier to wheelchair or pram users, even if a cable cover or mat is used.¹⁸⁶

These kinds of charging solution are quite new and many councils do not have a policy on whether, and under what circumstances, to permit them. The government published guidance to local highway authorities in December 2024 to assist them in developing such a policy, and the issues of pedestrian safety, ownership and maintenance they may have to consider.¹⁸⁷

In light of this guidance, many councils are still developing their policies on cross-pavement chargepoint permissions.¹⁸⁸

Funding for public chargepoints

The [Local Electric Vehicle Infrastructure](#) (LEVI) scheme is open to local authorities (in England only) and is designed to encourage large scale, ambitious and commercially innovative projects that aim to utilise more private sector investments.¹⁸⁹ The government said that £381 million had been committed to the LEVI scheme in May 2024.¹⁹⁰ The government has said it expects an additional 100,000 chargepoints across England to be installed,

¹⁸³ Merton Borough council, [Charging your electric vehicle](#), [accessed 22 May 2025]; Hertfordshire County Council, [Electric vehicle charging in Hertfordshire](#), [accessed 22 May 2025]. The Fast Charge, [106 days later: DfT yet to send councils pavement charging guidance](#), 16 January 2024

¹⁸⁴ [Highways Act 1980, s178](#)

¹⁸⁵ DfT, [Electric Vehicle Chargepoint Grant for Households with On-Street Parking](#), 18 March 2024

¹⁸⁶ Leicestershire County Council, [Electric vehicles \(EV\)](#), [Accessed 22 May 2025]

¹⁸⁷ DfT guidance, [Cross-pavement solutions for charging electric vehicles](#), 24 December 2024

¹⁸⁸ Hertfordshire County Council, [Electric vehicle charging in Hertfordshire](#), [accessed 22 May 2025]; Leicestershire County Council, [Electric vehicles \(EV\)](#), [Accessed 22 May 2025]

¹⁸⁹ Energy Saving Trust, [Local Electric Vehicle Infrastructure scheme](#), [online] accessed 16 February 2023

¹⁹⁰ PQ 25092 [[On Electric Vehicles: Charging Points](#)], answered 14 May 2024

using LEVI funding.¹⁹¹ Funding allocations to local authorities are listed online.¹⁹²

The now-closed [On-street Residential Chargepoint Scheme \(ORCS\)](#) provided grant funding for local authorities across the UK towards the cost of installing on-street residential chargepoints for plug-in electric vehicles. Grants provided up to 50% of the capital cost of installing a chargepoint (capped at £7,500 per chargepoint and £200,000 total).¹⁹³ The scheme is now closed to new applicants.¹⁹⁴

Rapid charging fund

Separately, in March 2022 the previous government pledged a £950 million rapid charging fund (RCF) to unlock current barriers to deployment of rapid chargepoints at motorway and major road service stations, and enable “provision where the commercial case will not add up.”¹⁹⁵ The RCF was intended to support the rollout of at least 6,000 high powered super-fast chargers across England’s motorways by 2035.¹⁹⁶ It was envisaged there would be at least six high powered chargepoints at each motorway service area by the end of 2023, but in January 2024 the RAC said this target had not been achieved.¹⁹⁷

So far the RCF is still not open to applications from motorway service station operators, but a £70 million pilot scheme was launched in 2023 and the government has also consulted on how the wider RCF scheme should be designed.¹⁹⁸

The Guardian reported in March 2025 that ministers were considering re-allocating RCF resources.¹⁹⁹ When asked in May 2025, the government did not commit to continuing the RCF project:

The market has changed significantly since the previous government announced the Rapid Charging Fund in 2020, including increased private sector investment in infrastructure, a changed regulatory environment, and improvements in battery technology. There are now over 5,250 open-access, rapid and ultra-rapid chargepoints within 1 mile of the Strategic Road Network (England’s motorways and major A roads).

¹⁹¹ PQ 31268 [[On Electric Vehicles: Charging Points](#)], answered 28 February 2025

¹⁹² Office for Zero Emission Vehicles, [Local Electric Vehicle Infrastructure \(LEVI\) funding amounts](#), updated 7 September 2023

¹⁹³ Office for Zero Emission Vehicles, [On-Street Residential Chargepoint Scheme guidance for local authorities](#), updated 20 October 2023

¹⁹⁴ Energy Saving Trust, [Local electric vehicle infrastructure fund](#) (Accessed 3 July 2024)

¹⁹⁵ DfT, [UK electric vehicle infrastructure strategy](#), 25 March 2022, p7

¹⁹⁶ DfT, [Tenfold expansion in chargepoints by 2030 as government drives EV revolution](#), 25 March 2022

¹⁹⁷ RAC Press Release, [Government fails to hit motorway services high-power EV charger target by end of 2023](#), 2 January 2024; see the Library insight [Going the distance: Are motorways ready for more electric vehicles?](#), 5 June 2024

¹⁹⁸ DfT Closed consultation, [Rapid charging fund: scheme design](#), 6 December 2023

¹⁹⁹ The Guardian, [Ministers may siphon off stalled £950m fund for motorway chargers](#), 24 March 2025

We will ensure taxpayers' money is used as efficiently as possible to make the transition to electric vehicles a success. The Rapid Charging Fund pilot was launched in 2023 to better understand where to target government support. We will apply learning from the pilot and continue working with industry to inform future policy to boost charging infrastructure on the Strategic Road Network.²⁰⁰

More information on the challenges of installing rapid chargers at service stations can be found in the Library 2023 insight, [Going the distance: Are motorways ready for more electric vehicles?](#)²⁰¹

Amending building regulations for EVs charging

In 2021 the government legislated to require EV chargepoints to be installed in a range of new build homes and other buildings, by amending the 2010 building regulations. The changes came into force for new building approvals from 15 June 2022 and delivered on a commitment in March 2022's Taking Charge Strategy.²⁰²

Under the amended regulations:

- New residential homes with onsite parking must have their own chargepoint.
- New residential buildings with parking in a covered carpark must have a chargepoint for each dwelling with a parking space.
- Buildings converted for residential use, and which have a car park, must have a chargepoint for each dwelling with a car parking space.
- Residential buildings undergoing major renovation works which have more than 10 car parking spaces must have a chargepoint for each dwelling and all other car parking spaces with cable routes.
- New non-residential buildings with more than 10 car parking spaces must have a minimum of one charging point, and cable routes for at least 20% of the remaining spaces.
- Mixed-use buildings must use the above regulations with calculations based on the number of car parking spaces designated for use by the respective classes of occupant.²⁰³

²⁰⁰ PQ 52653, [[On Rapid Charging Fund](#)], answered 21 May 2025

²⁰¹ Library insight, [Going the distance: Are motorways ready for more electric vehicles?](#) 5 June 2023

²⁰² [The Building Regulations etc. \(Amendment\) \(England\) \(No. 2\) Regulations 2021](#); DfT, [UK electric vehicle infrastructure strategy](#), 25 March 2022, p11

²⁰³ [The Building Regulations 2010](#), Schedule 1, Part S

The government has published guidance on how the amended building regulations should be implemented.²⁰⁴

Accessibility and convenience for charging

Accessibility and convenience of vehicle charging and refuelling has frequently been raised by consumers as a key concern in choosing to purchase and use an EV.²⁰⁵

Following a 2021 consultation, in 2023 the government delivered on a commitment in the 2022 EV Strategy and introduced regulations to improve the customer experience of on-street EV charging.²⁰⁶ The 2023 regulations affect public chargepoints with a power of 8 kilowatts or above, and require their operators to:

- allow contactless payment and payment roaming to enable drivers to use a single app or Radio Frequency Identification (RFID) card across multiple charge point networks
- open up their data to third-party apps to make it easier for drivers to locate chargepoints, check charging speeds and see whether they are working and available
- maintain 99% reliability for their network of rapid charge points, measured as an annual average
- create a 24/7 free to use staffed telephone helpline for all public charge points
- use pence per kilowatt hour (p/kWh) as the standard pricing metric for all public charge points.²⁰⁷

These regulations are in addition to 2017 regulations which require that:

- all publicly accessible standard or rapid chargepoints deployed or renewed since 17 November 2017 must at least offer a ‘Type 2’ recharging connector. This means the majority of charging stations across the UK and Europe have connectors to fit most cars;
- geographic location data of publicly accessible recharging points must be made freely available to anyone who wishes to access it;

²⁰⁴ Ministry of Housing, Communities and Local Government statutory guidance, [Infrastructure for charging electric vehicles: Approved Document S](#), updated 17 April 2023

²⁰⁵ DfT Consultation outcome, [The consumer experience at public electric vehicle chargepoints](#), updated 30 March 2023

²⁰⁶ DfT Consultation outcome, [The consumer experience at public electric vehicle chargepoints](#), updated 30 March 2023; [The Public Charge Point Regulations 2023](#); DfT, [New laws to make charging an electric vehicle easier and quicker](#), 25 October 2023

²⁰⁷ [The Public Charge Point Regulations 2023, Explanatory notes](#)

- chargepoints must be available to anyone for ‘ad-hoc access’, i.e. without contract or ongoing financial commitments to the electricity supplier; and
- chargepoints incorporate intelligent metering systems, that enables customers to record real-time information regarding their energy use.²⁰⁸

Accessible EV charging for disabled people

In 2022 the British Standards Institute (BSI) produced a Publicly Available Specification (PAS) for accessible charge points, called ‘PAS 1899 - Electric vehicles accessible charging specification’.²⁰⁹ The creation of this specification was sponsored by the charity Motability Foundation, and the UK Government’s Office for Zero Emission Vehicles (OZEV).

The PAS is intended to provide requirements and recommendations to public and private bodies involved in procuring and providing public chargepoints,

The PAS covers:

- the physical aspects of the environment surrounding fixed chargepoints (such as kerb height, ground type) and the placing of chargepoints;
- factors to be taken account of in the design and specification of accessible chargepoints and their more immediate surrounding areas (such as the height of chargepoint components, bollard spacing, interface tilt, colours used), and;
- information (physical and digital), signals and indicators provided.²¹⁰

Following the PAS is voluntary, but the government said in January 2024 that it is encouraging local authorities bidding for government LEVI funding to incorporate it into their procurement models.²¹¹

In October 2023 the Motability Foundation endorsed assessment schemes run by AccessAble (an accessibility consultancy) and BSI Assurance UK.²¹² Their assessment schemes will check which public EV chargepoints comply with the

²⁰⁸ [Alternative Fuel Infrastructure Regulations 2017](#); Office of Product Safety and Standards, [Guidance on The Alternative Fuels Infrastructure Regulations 2017](#), updated 30 June 2023

²⁰⁹ British Standards Institute, [Electric Vehicles Accessible Charging Specification - PAS 1899](#), October 2022; [Office for Zero Emission Vehicles](#), About Us (Accessed 4 July 2024); Motability Foundation, [Accessible Electric Vehicle Charging](#) (Accessed 4 July 2024)

²¹⁰ British Standards Institute, [Electric Vehicles Accessible Charging Specification - PAS 1899](#), October 2022

²¹¹ [PQ 10558](#) [On [Electric Vehicles: Charging Points](#)] answered 25 January 2024

²¹² Motability Foundation, [Motability Foundation endorses two EV charge point assessment schemes](#), 16 October 2023; AccessAble, [About Us](#) (Accessed 4 July 2024); BSI, [About Us](#) (Accessed 4 July 2024)

PAS. Assessments will be made available online, so that people with accessibility needs know which chargepoints they can use.²¹³

The Public Accounts Committee (PAC)’s 2025 inquiry into [Public charge points for electric vehicles](#) found that “there are still no charge points in the UK which are fully compliant with the [PAS] standard” and that “Failure to address problems with uptake of the standard quickly will mean that the network will continue to develop without meeting the needs of drivers with disabilities.” It called on the government to set the “actions it is taking to ensure that charge point accessibility is improved”.²¹⁴ The government is yet to respond.

Devolved administrations

Local transport, strategic planning, land use and public transport subsidies are all devolved competencies.²¹⁵ Therefore the Scottish, Welsh and Northern Ireland administrations each have their own EV chargepoint and infrastructure strategies.²¹⁶

2.5

Decarbonising the bus fleet

The Conservative government committed £525 million to support the transition to Zero-Emission Buses (ZEBs) and achieve its target of having 4,000 ZEBs on the road by the end of the previous Parliament.²¹⁷ Most ZEBs are battery electric buses, and a much smaller proportion are hydrogen fuel cell buses (see table below).

In September 2023 the government said it had reached its target of funding at least 4,000 ZEBs.²¹⁸ In May 2024 the then Secretary of State for Transport, Mark Harper, said “more than 5,200 buses have been funded across the UK since February 2020, with UK bus manufacturers supporting many of them.”²¹⁹ However, the number of ZEBs currently on the road is likely to be lower than

²¹³ Motability Foundation, [Motability Foundation endorses two EV charge point assessment schemes](#), 16 October 2023

²¹⁴ Committee of Public Accounts, [Public charge points for electric vehicles](#), 2024–25 HC 512, 12 March 2025

²¹⁵ DfT, [UK electric vehicle infrastructure strategy](#), 25 March 2022, p31

²¹⁶ Transport Scotland, [Vision for world class public electric vehicle charging network](#), 18 June 2023; Welsh Government, [Electric vehicle charging: strategy and reports](#), updated 3 July 2023; Northern Ireland Department for Infrastructure, [Action Plan for Electric Vehicle Infrastructure](#), 25 November 2022

²¹⁷ Transport Select Committee, [Implementation of the National Bus Strategy](#), 30 March 2023, HC 161 2022–23, para 90; Prime Minister’s Office press release, [Major boost for bus services as PM outlines new vision for local transport](#), 10 February 2020

²¹⁸ DfT Press Release, [Hundreds of new zero emission buses to connect communities in England](#), 8 September 2023

²¹⁹ [HC Deb 16 May 2024 c410](#)

this. As of June 2024 there were 4,086 licensed battery electric buses and coaches in the UK (see table below).²²⁰

The main funding source for ZEBs in England is the Zero Emission Bus Regional Areas (ZEBRA) scheme which was available to local authorities in England outside of London but has now closed. Over £400 million has been allocated through the scheme, and allocations by local authority are listed online.²²¹ The ZEBRA scheme follows the Ultra-Low Emission Bus Scheme (which ran in 2018), and the Low Emission Bus Scheme (LEBS, first announced in 2015).²²² In November 2024 the government said that of the 2,263 buses funded through the ZEBRA 1 scheme, 1,674 buses had been ordered and 582 ZEBs were in service.²²³

The ability for the UK Government to meet its target is also partially dependent on actions taken by the devolved administrations and their respective ZEB funding streams.²²⁴

Manufacturers of battery electric buses

The UK has three main bus manufacturers: Alexander Dennis, Switch Mobility, formerly known as Optare, and Wrightbus. They have all developed electric buses and have a number in service across the UK. As of 2023, the three companies combined employ over 3,500 people.²²⁵

The table below shows the number of electric buses and coaches produced by different manufacturers. It shows that 37% (1,506 vehicles) were produced by British companies.

The manufacturers with the greatest proportion of the UK battery electric bus fleet was BYD (41%), followed by Wrightbus (25%) and Yutong (11%). BYD and Yutong are Chinese companies.

²²⁰ Department for Transport (DfT), [Vehicle licensing statistics data tables](#), VEH0105

²²¹ DfT Transparency data, [Zero Emission Bus Regional Areas \(ZEBRA\) programmes funding amounts](#), updated 22 March 2024

²²² DfT Transparency data [Low emission bus scheme: successful bidders](#) 28 August 2017; DfT Transparency data [Ultra-low emission bus scheme: successful bidders](#) 6 February 2019

²²³ PQ 15317 [[on Buses: Carbon Emissions](#)], answered 29 November 2024

²²⁴ Transport Select Committee, [Implementation of the National Bus Strategy](#), 30 March 2023, HC 161 2022–23, para 89; Transport Scotland, [Scottish Ultra-Low Emission Bus Scheme](#), (accessed 16 May 2024); Transport Scotland, [Scottish Zero Emission Bus challenge fund](#), (accessed 16 May 2024)

²²⁵ Alexander Dennis Limited, [Annual report and financial statement for the year ending 1 January 2023 \(PDF\)](#); Switch Mobility Limited, [Annual report and financial statement for the year ending 31 March 2023 \(PDF\)](#); Wrightbus press release, [Hundreds of new jobs at Wrightbus](#), 30 August 2023.

Licensed battery electric buses and coaches in the UK by make at the end of June 2024

Vehicle make	Number of vehicles	Percentage of vehicles
UK based	1,506	37%
WRIGHTBUS	1,002	25%
OTHER BRITISH	244	6%
OPTARE	164	4%
SWITCH	62	2%
ALEXANDER DENNIS	34	1%
Other countries	2,580	63%
BYD	1,671	41%
YUTONG	469	11%
VOLVO	156	4%
FIAT	56	1%
LDV	47	1%
CAETANO	43	1%
FORD	40	1%
Other	98	2%
Total	4,086	100%

Source: Department for Transport (DfT), [Vehicle licensing statistics data tables](#), VEH01333b (GenModels)

In a Commons debate in May 2024, MPs raised concerns that Chinese manufacturers were winning many UK bus contracts over UK manufacturers.²²⁶ In response the then Transport Secretary Mark Harper said that trade rules prevented the Government from instructing bus operators or local authorities from buying ZEBs from British manufacturers, but that the Government was supporting the industry:

it is not possible, given our international commitments under the World Trade Organisation, to specify that people have to buy British buses. [...] British bus manufacturers are very competitive. The Government have made support available to businesses through our Advanced Propulsion Centre and UK Export Finance. As I said to [Alan Brown], if the shadow Minister thinks that there is any unfair competition with subsidised imports, the Trade Remedies Authority has all the tools at its disposal to deal with that.²²⁷

Proportion of local bus operators using electric buses

The table below shows that at the end of March 2024, 8.1% of buses used by local bus operators in Great Britain were electric. This compares to 73.1% that were diesel and 13.1% that were diesel hybrid. London had the highest

²²⁶ [HC Deb 16 May 2024 c410](#)

²²⁷ [HC Deb 16 May 2024 c411](#)

proportion of electric buses (16.7%), followed by Scotland (14.0%) and then Wales (8.0%).²²⁸

Percentage of buses used as Public Service Vehicles by fuel type in 2023/24

	Electric (not hybrid)	Diesel- Hybrid	Diesel	Hydrogen	Methane/ Biomethane	Other/ Unknown
London	16.7%	42.8%	39.3%	0.3%	0.0%	1.0%
English metropolitan areas	4.5%	6.9%	82.8%	0.4%	0.4%	5.0%
English non-metropolitan	3.1%	1.1%	88.5%	0.1%	2.5%	4.7%
England	7.4%	14.8%	72.7%	0.2%	1.2%	3.7%
Scotland	14.0%	3.3%	71.1%	0.4%	0.0%	11.1%
Wales	8.0%	0.0%	86.9%	0.0%	0.0%	5.1%
England outside London	3.6%	3.2%	86.5%	0.2%	1.7%	4.8%
Great Britain	8.1%	13.1%	73.1%	0.3%	1.0%	4.5%

Source: Department for Transport (DfT), [Bus statistics data tables](#), Vehicles operated by local bus operators (BUS06e)

2.6 Decarbonising freight

Heavy Goods Vehicles (HGVs) account for 16% and light vans account for 16% of domestic transport emissions despite only accounting for 5% and 17% of total vehicle mileage. In comparison, cars and taxis currently account for 75% of mileage while creating just over half (54%) of transport emissions.²²⁹

The government said in May 2025 that “more needs to be done to decarbonise the road freight sector” and it was “considering [its] regulatory approach for decarbonising new non-zero emission heavy goods vehicles and look[ed] forward to engaging the sector on this in due course.”²³⁰

Future of freight plan

The Department for Transport (DfT) published its [Future of Freight plan](#) on 15 June 2022. This plan sets out a vision for a freight sector that is “cost-efficient, reliable, resilient, environmentally sustainable and valued by society”.²³¹

²²⁸ Department for Transport (DfT), [Bus statistics data tables](#), Vehicles operated by local bus operators (BUS06)

²²⁹ (DfT), [Energy and environment: data tables](#), ENV0201; 12 December 2024; DfT, [Road Traffic Estimates](#), TRA101

²³⁰ PQ 52652 [[Large Goods Vehicles: Carbon Emissions](#)], answered 21 May 2025

²³¹ DfT, [Future of Freight](#), 15 June 2022, para 4.12

The Future of Freight plan restates the Government's commitment to Net Zero emissions by 2050.²³² It also restates the phase-out dates of 2035 for non-zero emission HGVs weighing 26 tonnes (with some potential exemptions), and 2040 for all non-zero emission HGVs.²³³

In 2023 the SMMT said that meeting these targets will require an urgent government led-strategy, especially on the installation of public HGV-focussed battery-charging and hydrogen provision which are currently non-existent on England's major roads and motorways, meaning very few zero-emission HGVs were actually being bought and used:

The heavy goods vehicle (HGV) sector is today calling for an urgent, government-led strategy to drive uptake of zero emission trucks to meet decarbonisation targets. With just over a decade until the first end of sale deadline for the UK's predominantly diesel HGV workhorses, [our] analysis... shows that Britain's strategic road network has not a single HGV-dedicated electric charging or hydrogen filling point. This lack of infrastructure makes it impossible for the vast majority of operators to contemplate investments to decarbonise their fleets. [...]

HGV manufacturers are investing heavily in a broad range of electric and hydrogen models, however, these currently account for just one in 600 trucks on UK roads. A key concern for the sector is 'charging anxiety', which is felt acutely by operators with time-sensitive, long-haul business models.²³⁴

In May 2024 the SMMT said that, in the first quarter of 2024, battery-electric trucks accounted for just 0.5% of new HGV sales.²³⁵

The government's Future of Freight plan recognises that there is uncertainty in the sector, particularly for heavier and longer haul freight, about what fuels and technology are required for the sector to reduce its emissions. This uncertainty, coupled with the long life-span and high cost of freight assets, and concerns about the risk of "stranded assets and first-mover disadvantages" are creating barriers to private investment in new technologies.²³⁶

Zero Emission HGV and Infrastructure Demonstrator Programme

In 2022 the government announced it would invest £200 million into research projects aimed at exploring how the sector could decarbonise.²³⁷ Four 'zero emission HGV and infrastructure demonstrator programme' research projects

²³² [Section 1 of the Climate Change Act 2008](#)

²³³ DfT, [Future of Freight](#), 15 June 2022, p117

²³⁴ SMMT, [Zero Emission HGV market shackled by absence of infrastructure and lack of plan](#), 18 April 2023

²³⁵ SMMT, [HGV growth stabilises while zero emission market share rises](#), 16 May 2024

²³⁶ DfT, [Future of Freight](#), 15 June 2022, p54-58

²³⁷ DfT Press release, [£200 million boost to rollout of hundreds more zero-emission HGVs](#), 12 May 2022

were awarded funding in 2023. These projects are looking into battery electric trucks, hydrogen, fuel cell technologies, or combinations of these.²³⁸

The government said the demonstrator projects will “roll out up to 370 zero emission HGVs” and deliver “around 57 refuelling and electric charging sites, providing the crucial infrastructure to help the haulage sector decarbonise.”²³⁹

Towards a strategy on zero emission HGV and coach infrastructure

In October 2023 the government ran a call for evidence on infrastructure for zero emission heavy goods vehicles and coaches, which sought evidence:

- to support the development of a zero emission HGV and coach infrastructure strategy for the UK
- regarding the zero emission HGV and coach markets in the UK, including its infrastructure, both public and private (for example, depot-based)
- to inform future decision-making about zero emission HGVs and coaches.²⁴⁰

The call for evidence confirmed that the government is “technology neutral” and it was open to different ways of decarbonising vehicles, including HGVs:

A zero emission vehicle is a vehicle that does not produce harmful exhaust emissions, such as greenhouse gas and pollutant emissions. Exhaust emissions that do not damage public health, such as water vapour, are permitted.

UK government policy on the transition to zero emissions is technology neutral. Both battery electric and hydrogen fuel cell, as zero emission technologies, will have a role in the decarbonisation of the HGV and coach sectors.²⁴¹

Following the 2023 call for evidence, in January 2024 the government said it planned to launch a strategy on zero emission HGV and coach infrastructure “in early 2024”.²⁴² Such a strategy was not launched by the end of the last Parliament.

Coaches

In terms of decarbonisation coaches are usually grouped with HGVs rather than with buses, because of the longer distances they cover. Further detail on

²³⁸ DfT, [Zero emission heavy goods vehicles and infrastructure competition winners](#), 24 October 2023

²³⁹ DfT Press release, [Government invests £200 million to drive innovation and get more zero emission trucks on our roads](#), 19 October 2023;

²⁴⁰ DfT, Closed call for evidence, [Infrastructure for zero emission heavy goods vehicles and coaches](#), 19 October 2023

²⁴¹ DfT, Closed call for evidence, [Infrastructure for zero emission heavy goods vehicles and coaches](#), 19 October 2023

²⁴² PQ 8218 [[On Large Goods Vehicles: Electrification](#)], answered 11 January 2024

[the challenges of decarbonising the coach fleet can be found in a report \[PDF\]](#) on the issue by the Confederation Of Passenger Transport (CPT), the industry body representing bus and coach companies.²⁴³

The government said in May 2025 that it expected the coach sector to opt for zero-emission vehicles on commercial grounds, and did not plan to publish a roadmap for coach decarbonisation:

We have no current plans to draft a roadmap. The Government is working with the sector to fully decarbonise the coach fleet. We are pleased to see the progress that has been made so far, particularly as long distance zero emission coaches have only been on the market since 2024. We expect to see the numbers of zero emission coaches on UK roads continue to grow as more coach operators opt for them on commercial grounds. We also welcome that some coach operators are starting to use charging infrastructure at bus depots and strongly encourage this shared use of facilities.²⁴⁴

²⁴³ Confederation Of Passenger Transport, [What are the Challenges to Transitioning to a Zero Emission Coach Fleet?](#), 2022

²⁴⁴ PQ 53313 [[On Bus Services: Carbon Emissions](#)], answered 27 May 2025

3

The integration of EVs into the energy system

Increasing the number of EVs will reduce petrol and diesel demand but add to electricity demand. This could increase pressure on Great Britain's electricity grid network, which is operated by the [National Energy System Operator](#) (NESO).²⁴⁵

3.1

Concerns about electric vehicles exceeding electricity demand

Following the government's initial announcement in July 2017 of plans to ban the sale of new conventional petrol and diesel cars and vans from 2040, concerns were raised by the media that this policy would require significantly more capacity in the power sector and present challenges for balancing the electricity grid.²⁴⁶

In response to concerns, National Grid published a [myth-buster on electric vehicles](#) (updated in September 2023), which stated that:

It's important to remember that this is a managed transition; the shift to EVs is happening gradually – not overnight. In England and Wales [...] the growth in offshore wind farms means that wind power will adequately meet the future demand for electrifying transport.²⁴⁷

It also estimated that even if all motorists switched to EVs overnight, electricity demand would increase by around 10%, which would be “well within the range the grid can capably handle”.²⁴⁸

When in 2020 the government brought forward, the ban on sales of new petrol and diesel cars and vans to 2030 (see section 2.1 of this briefing), the potential difficulties for the electricity grid were no longer the focus of media

²⁴⁵ NESO is an independent, public corporation responsible for planning Britain's electricity and gas networks and operating the electricity system.

²⁴⁶ The 2040 target date was first announced by Defra and DfT in July 2017 ([Plan for roadside NO2 concentrations published](#), July 2017), see section 2.1 of this briefing for information on targets; [Diesel and petrol car ban: Plan for 2040 unravels as 10 new power stations needed to cope with electric revolution](#), The Daily Telegraph, 27 July 2017.

²⁴⁷ National Grid, [Busting the myths and misconceptions about electric vehicles](#), updated September 2023.

²⁴⁸ National Grid, [Busting the myths and misconceptions about electric vehicles](#), updated September 2023.

concerns.²⁴⁹ National Grid reportedly said the grid was “suitably robust to cope” with increased EV demand from 2030.²⁵⁰

The House of Lords Environment and Climate Change Committee, in its 2024 report on [EV strategy: rapid recharge needed](#), also noted that witnesses had told them that “the transition to EVs would not impose significant additional demand on the grid”.²⁵¹

3.2

The impact of electric vehicles on energy demand

NESO publishes an annual Future Energy Scenarios (FES) report presenting different ways to decarbonise the UK energy system. The [2022 FES report](#) provided four scenarios, which differed in the speed of decarbonisation, the extent of consumer behaviour and the mix of technologies used.²⁵²

The 2022 FES found that total energy demand for road transport would fall across all scenarios as a result of the switch to electric vehicles but that this would lead to an increase in electricity demand:

Electrification of transport leads to increased annual electricity demands in all scenarios, but lower overall energy demand as electricity displaces petrol and diesel. Electric Vehicles have much higher efficiency and travel further on one kWh of electricity than one kWh of petrol, so total energy demand reduces. Overall transport demand is lowest in scenarios with the most electrification of transport.²⁵³

Each of the scenarios in the 2022 FES predict a fall in transport demand from around 400 TWh in 2024 to around 150-120 TWh in 2050, with electricity meeting around 100-120 TWh of that demand in 2050 up from negligible levels in 2024. The government’s March 2022 [UK electric vehicle infrastructure strategy](#) also forecast an increase in electricity demand, estimating that

²⁴⁹ BEIS, Prime Minister’s Office, 10 Downing Street, The Rt Hon Sir Alok Sharma KCMG MP, and The Rt Hon Boris Johnson MP, [UK sets ambitious new climate target ahead of UN Summit](#), December 2020. Commentators critical of the 2030 target instead highlighted the challenges of the earlier date for manufacturers of conventional petrol and diesel vehicles, and the potential cost and choice impacts on motorists. See for example: The Guardian, [Vauxhall plant at risk after ‘brutal’ 2030 ban, says Stellantis boss](#), 19 January 2021; The Telegraph [Banning new petrol and diesel cars by 2030 is feasible but don’t penalise the auto industry](#), 18 November 2020; The Daily Mail, [‘Driving will become a privilege for the wealthy’ \[...\]](#), 17 November 2020.

²⁵⁰ This is Money, [National Grid confident it is ‘suitably robust to cope’ with increased electric vehicle demand from 2030 when new petrol and diesel cars will be banned](#), 17 November 2020.

²⁵¹ Lords E&CC Committee, [EV strategy: rapid recharge needed](#), 1st Report of Session 2023-24, para 276

²⁵² NESO published a [2024 Future Energy Scenarios \(PDF\)](#) report in July 2024, providing new scenarios and modelling. The broad conclusions referred to in this section of the paper remain similar.

²⁵³ National Grid ESO, [Future Energy Scenarios 2022](#) [PDF], July 2022, p68.

“road transport could represent approximately 15% of total electricity demand in 2050 compared to less than 1% of demand today”.²⁵⁴

3.3

Balancing electricity supply and demand

The supply of electricity must match demand at all times in order to prevent power cuts or spikes in network frequency that could damage equipment. This is known as ‘balancing’ and is carried out by NESO.

As shown above, wider proliferation of EVs will create an additional source of demand to the grid that will need to be balanced. However, two approaches – smart charging and vehicle-to-grid – could help to manage this increase in demand by reducing charging at peak times or enabling the batteries in EVs to become an asset to NESO to assist in grid balancing.

For more information on grids, including how they are structured and balanced, see the Library briefing paper on [Electricity grids](#)

Smart charging

‘Smart’ use of the electricity system involves shifting the use of electricity, where possible, to times when demand is low. Consumers can benefit from cheaper power, and operators benefit by offsetting some of the additional demand from EVs to times of low demand or high levels of supply from renewables.

Smart charging requires a customer to have a smart meter. These meters have been rolled out for several years²⁵⁵ and have the potential to allow more detailed information on consumption to be sent to energy suppliers. They also allow customers to react to price signals in the power system. For example, ‘time-of-use’ tariffs are already available from some energy suppliers, allowing customers with smart meters to pay lower unit prices when using power at times of low demand, including to charge their EV.²⁵⁶

NESO’s new [Demand Flexibility Service](#) (DFS) operates on similar principles to time-of-use tariffs. It has rewarded customers with payments for reducing energy consumption during a small number of ‘test periods’ of very high demand over winter 2022-23.²⁵⁷ The DFS ran again in 2023/24. From November 2024 NESO has continued to run the DFS as a measure that is now available all-year round rather than just during specific ‘test’ periods in winter. The DFS

²⁵⁴ HM Government, [Taking charge: the electric vehicle infrastructure strategy](#) (PDF), March 2022, p121.

²⁵⁵ For more information on smart meters see the Library briefing paper, [Energy Smart Meters. The Energy Act 2023](#) extended the powers for the smart meter roll out which were due to expire in 2023 to 20228. For more information on the bill’s smart meter provisions see BEIS, [Energy Security Bill factsheet: Smart metering](#), 1 September 2023.

²⁵⁶ Energy Saving Trust, [Time of use tariffs: all you need to know](#), updated 11 June 2024.

²⁵⁷ NESO, [Demand flexibility service explained](#) [accessed 21 May 2025].

now competes with other means of supplying capacity to the grid, meaning prices are typically lower.²⁵⁸

The government and Ofgem published the Smart Systems and Flexibility Plan (SSFP) in July 2021, which envisioned that by the mid-2020s “drivers will choose to smart charge because it is convenient and economical” and that by 2030 smart charging will be the norm.²⁵⁹ However, NESO’s [Future Energy Scenarios 2023](#) suggests that there is a degree of uncertainty relating to consumer engagement with smart charging, with the amount of reduction in peak demand increasing as the penetration of smart meters and time-of-use tariffs increases. It states that:

While we expect the majority of those who can charge at home to do, their charging behaviour will depend on the technology, incentives and level of automation that is in place. Nevertheless, we have seen early adopters of smart charging respond to new services such as the ESO’s Demand Flexibility Service over winter 22/23.²⁶⁰

NESO’s [2024 Future Energy Scenarios report](#) recommended actions to increase consumer engagement with smart charging, including giving consumers more information on the benefits and introducing competitive, nationwide off-peak charging rates for public and kerbside charge points. It noted that smart charging needs to become “effortless for consumers, featuring autonomous functionality”.²⁶¹

Vehicle-to-grid (V2G)

An extension of smart charging, the concept of ‘Vehicle to Grid’ (V2G or V2X which means ‘Vehicle-to-everything’), is that when supply is low and demand high, EVs connected to the grid to charge can instead release power back into the grid. Owners of the vehicles can then be paid for this balancing service in a similar way to electricity storage operators. National Grid ESO estimates that “by 2050, vehicle-to-grid could offer between 8-39 GW for the electricity market in Great Britain (representing 7–17 % of total [balancing])”.²⁶²

Some energy suppliers have been developing V2G offers for their customers.²⁶³ However, National Grid ESO’s [Future Energy Scenarios 2023](#) notes that engagement is expected to be low in the short term:

²⁵⁸ NESO, [Demand flexibility service explained](#) [accessed 21 May 2025]; Ofgem, [Decision on the Demand Flexibility Service in relation to an update to the Terms and Conditions related to balancing](#) (PDF), 21 November 2024.

²⁵⁹ BEIS & Ofgem, [Transitioning to a net zero energy system: Smart Systems and Flexibility Plan 2021](#) [PDF], July 2021, p19.

²⁶⁰ National Grid ESO, [Future Energy Scenarios 2023](#), July 2023.

²⁶¹ NESO, [Future Energy Scenarios 2024](#) (PDF), July 2024, page 62.

²⁶² NG ESO, [Future Energy Scenarios 2021](#) [PDF], July 2021, cited in BEIS & Ofgem [Networks Strategic Framework: Enabling a secure, net zero energy system](#) [PDF], August 2022, pp31-32.

²⁶³ For example, [Octopus energy’s Powerloop product](#), (Accessed 10 July 2024); More information is available in the Library briefing paper on [Electricity Grids](#) (Section 5 – smart grids).

Vehicle-to-Grid engagement is expected to remain very low in the short-term. Increased take-up will require technology deployment of e.g. bi-directional charging, but also electricity market change so consumers are rewarded for participating in the energy system to optimise system costs for everyone.²⁶⁴

NESO's [2024 Future Energy Scenarios report](#) said that “more new vehicles need to have V2G functionality to empower consumers to engage with the process” and more communications are needed with consumers on the benefits.²⁶⁵ It also recommended that financial support was needed to help consumers participate at scale.

The House of Lords Environment and Climate Change Committee, in its report on [EV strategy: rapid recharge needed](#), discussed both smart charging and V2G technologies. It found that although both approaches had merits, barriers still existed that were holding up deployment and recommended the government explore what more can be done to integrate the technology.²⁶⁶ The government agreed with this recommendation and in its response to the enquiry noted that:

As part of the Electric Vehicle Smart Charging Plan 2023, the Government committed to investigating the technical and economic potential of smart charging in public places. The Government is continuing work to address barriers to smart charging at all long stay public locations.²⁶⁷

3.4

Policies to support smart charging and V2G

In March 2022 the Department for Transport published the [EV Infrastructure Strategy](#), setting out plans for the rollout of electric vehicle charging infrastructure.²⁶⁸ The report explained in more detail how the government and Ofgem are working with energy companies to develop solutions to manage the increased demand from electrical vehicles, including “a systems-wide approach for a safe and secure transition to smart charging.”²⁶⁹

Other recent government policies included:

- A £12.6 million [V2X innovation programme](#), which aims to address barriers to bidirectional electric vehicle charging.²⁷⁰ This followed the award of up to £30 million to fund 21 V2G projects over 2018-2022, as part of an expanded £20 million competition launched jointly by the

²⁶⁴ National Grid ESO, [Future Energy Scenarios 2023](#), July 2023.

²⁶⁵ NESO, [Future Energy Scenarios 2024](#) (PDF), July 2024, page 63.

²⁶⁶ Lords E&CC Committee, [EV strategy: rapid recharge needed](#), 1st Report of Session 2023-24, para 253.

²⁶⁷ HM Government, [Government response to the Environment and Climate Change Committee report 'EV Strategy: rapid recharge needed'](#) [PDF], April 2024.

²⁶⁸ DfT, [UK electric vehicle infrastructure strategy](#), 25 March 2022.

²⁶⁹ DfT, [UK electric vehicle infrastructure strategy](#), 25 March 2022, p81.

²⁷⁰ BEIS, [V2X Innovation Programme](#), 17 January 2023.

Department for Business Energy and Industrial Strategy (BEIS) and the Office for Low Emission Vehicles in 2017.²⁷¹

- The [Electric Vehicles \(Smart Charge Points\) Regulations 2021](#) that came into force on 30 June 2022. They require electric vehicle charge points sold in Great Britain for domestic or workplace use to have smart functionality, and a default off-peak setting to encourage electrical vehicle owners to charge at times of lower demand (and cost).²⁷²
- The [electric vehicle smart charging action plan \(January 2023\)](#), which set out plans to “deliver energy flexibility from electric vehicles, and to make sure the system is ready to respond in time for the upturn in energy demand”. These include: delivering a V2X Innovation Programme to address barriers to deployment by 2025; working with DNOs to ensure there is a consistent connection process for V2X across all regions; and ensuring all private charge points are interoperable with DSR services providers.²⁷³ The action plan builds on the 2021 [Smart Systems and Flexibility Plan](#) and responses to BEIS’ 2021 consultation on the system²⁷⁴
- The [Energy Act 2023](#), which brought in provisions that allow the Secretary of State to make ‘energy smart regulations’ for electric vehicle charge points to enable smart charging.²⁷⁵ The government [consulted on proposed regulations](#) between April and June 2024. The government’s response was published in April 2025, and said the government would bring forward new regulations “within the next year”.²⁷⁶

3.5 Grid connections

Concerns have also been raised about upgrades to the GB electrical grid system required to accommodate EVs and the related charging infrastructure. This includes allowing charge points to be physically connected to the grid, such as in the case of motorway services that report

²⁷¹ DfT, [£30 million investment in revolutionary V2G technologies](#), Feb 2018; DfT press notice, [Innovative vehicle to grid technology to receive £20 million](#), 8 July 2017. In 2023, BEIS was split into the Department for Energy Security and Net Zero, the Department for Business and Trade and Department for Science, Technology and Innovation.

²⁷² [Electric Vehicles \(Smart Charge Points\) Regulations 2021 SI 2021/1467](#). The Government guidance [Regulations: electric vehicle smart charge points](#) (27 October 2022) provides an overview of the regulations.

²⁷³ BEIS & Ofgem, [Electric vehicle smart charging action plan](#), January 2023.

²⁷⁴ BEIS, [Role of vehicle-to-X energy technologies in a net zero energy system: call for evidence](#), 20 July 2021.

²⁷⁵ Further details of the provisions in the Act that relate to smart charging and load control can be found in Part 8 of the Library briefing [Energy Bill \[HL\] 2022-23, parts 7-10: heat networks, smart appliances, load control and energy performance of buildings](#) (May 2023).

²⁷⁶ DESNZ, [Delivering a smart and secure electricity system: implementation](#), 23 April 2025.

difficulties in procuring sufficient grid capacity to upgrade to rapid chargers.²⁷⁷

A [report in August 2023 by the UK's Electricity Networks Commissioner, Nick Winser](#), highlighted the need to speed up the development of new electricity transmission infrastructure in order to connect the new generating capacity required to electrify transport and other demands.²⁷⁸ Transmission upgrades are currently taking around 12 to 14 years. NESO is leading an ongoing programme of reform to the grid connections process.²⁷⁹

In December 2024 the Office for Zero Emission Vehicles published a policy paper on “[Improving the grid connection process for electric vehicle charging infrastructure](#)” that outlined a number of action areas for improving grid connections. This included topics such as streamlining connection processes with Distribution Network Operators, standing charges for EV charge points and wayleaves/land access for charge points.

²⁷⁷ Forecourt Trader, [Motorway services operators demand “national priority” to access grid power for EV charging](#), January 2024.

²⁷⁸ DESNZ, [Accelerating electricity transmission network deployment: Electricity Networks Commissioner's recommendations](#), August 2023.

²⁷⁹ The Library debate pack, [Delivery of electricity grid upgrades](#) (November 2024) and [briefing on the Planning and Infrastructure Bill](#) (section 3.3, March 2025) provide background. In April 2025, [Ofgem approved NESO's plans to reform the connections process](#). This will prioritise connections for projects that are “[ready to progress, and which are needed to deliver clean power by 2030](#)”.

4 Environmental impact: EV lifecycle emissions and battery decommissioning

EVs are promoted as a better environmental option than conventional internal combustion vehicles.²⁸⁰ However there has been some debate over environmental concerns with EVs, including their lifecycle emissions, the manufacturing emissions associated with producing a battery using critical minerals, and decommissioning EV batteries at the end of their operating life.²⁸¹

4.1 Lifecycle emissions

The total emissions from electric vehicles are known as the “lifecycle emissions”. These combine the emissions from:

- manufacturing the vehicle: which tend to be higher than internal combustion engine vehicles and includes the mining of critical minerals (see Box below),
- powering the vehicle through its life: which tend to be lower than internal combustion engine vehicles though this varies on the electricity mix the EV is charged with,
- decommissioning the batteries at the end of their life.

Several studies have been conducted on EV lifecycle emissions with varying conclusions based on the methodology and assumptions used by the researchers. These include factors such as the size of the car, driving and efficiency assumptions, and where the car is manufactured and charged. In general, these show that EVs tend to have lower emissions over their lifecycle than traditional vehicles, with the lower emissions from powering the vehicle making up for the higher manufacturing emissions after a few years.²⁸²

The International Energy Agency (IEA) has created a [calculator which provides lifecycle emission comparisons](#) between vehicles based on factors such as their size and their country of purchase. For a medium size car in the

²⁸⁰ Local Government Association, [The case for electric vehicles](#) (accessed 10 July 2024).

²⁸¹ See for example, Simon Evans, [Fact check: why Rowan Atkinson is wrong about electric vehicles](#), The Guardian, 8 June 2023.

²⁸² DfT, [Lifecycle analysis of UK road vehicles](#), 25 March 2022.

UK in June 2024 this states emission savings of up to 58% for a battery EV compared to an ICE equivalent:

A typical medium car with a petrol (gasoline) engine bought in the United Kingdom and driven 33 km per day will be responsible for life-cycle emissions of 28.4 tonnes of CO₂ equivalent over a 15-year lifetime in the Stated Policies scenario.

An equivalent plug-in hybrid EV would produce 21.7 tonnes, or 24% less over its lifetime.

An equivalent battery EV with a 300 km range would produce 11.9 tonnes, 58% less over its lifetime than a conventional internal-combustion vehicle and 45% less than an equivalent plug-in hybrid EV. Despite higher manufacturing emissions associated with producing the battery, the battery EV's cumulative emissions are lower than those of its internal-combustion equivalent after 3 years.²⁸³

As such, while EVs are not technically 'zero emission', evidence suggests that in the UK they are likely to have fewer emissions than the average conventional vehicle over a lifetime of use. EV emissions have the potential to be reduced further in future as the power sector decarbonises and if manufacturing emissions are reduced.

With regards to UK emissions, the transport sector is now the largest sector of emissions having overtaken the energy sector in 2016.²⁸⁴ In 2021 the energy sector produced 20% of UK emissions, compared to 26% from transport.²⁸⁵ The decarbonisation of the energy sector (specifically power) will help lower the lifecycle emissions of EVs, and, alongside increased car efficiency, other low carbon fuels, and lower manufacturing emissions, increases in EVs vehicles will help to decarbonise the transport sector.

²⁸³ IEA, [EV Life Cycle Assessment Calculator](#), 5 June 2024.

²⁸⁴ BEIS, [2016 UK Greenhouse gas emissions, final figures](#), 6 February 2018 (PDF).

²⁸⁵ BEIS, [Final UK greenhouse gas emissions national statistics: 1990 to 2021](#).

Critical minerals in EV batteries

Batteries for electric vehicles can require rare elements such as lithium and cobalt, which has raised environmental and ethical issues in countries where these elements are mined and processed into batteries as well as questions over sustainable and secure supply as demand for batteries grows.²⁸⁶ The previous Conservative government published a [UK critical minerals strategy](#) in July 2022, and a [Critical minerals refresh](#) in March 2023, which set out the then government's plans to improve the resilience of critical minerals supply chains. Critical minerals are those which are important for the economy or national security or have a high risk of supply disruption.²⁸⁷

In 2025 the government commissioned an [analysis of UK critical minerals capabilities](#).²⁸⁸ The government has said that the Department for Business and Trade will publish a new critical minerals strategy in 2025.²⁸⁹

For further information, see:

- Library debate pack, [Supply of lithium and other critical minerals](#), April 2023
- POST briefing, [Mining and the sustainability of metals](#), February 2022
- POSTnote, [Access to Critical Materials](#), September 2019.

4.2 End of life for electric vehicle batteries

The number of EVs currently approaching the end of their life is relatively small, however the EV market is growing which in turn will lead to more batteries reaching the end of their on-vehicle life. For example, in 2023 the McKinsey consultancy estimated that battery recycling will grow as demand for lithium ion batteries grows (from about 700 GWh in 2022 to 4.7 TWh in 2030) and more batteries reach their end of life:

Although the recycling segment is expected to be relatively small in 2030, it is projected to grow more than three-fold in the following decade, when more batteries reach their end-of-life.²⁹⁰

²⁸⁶ POST briefing, [Mining and the sustainability of metals](#), February 2022.

²⁸⁷ The [UK Critical Minerals Intelligence Centre](#) publishes analysis of critical minerals supply and in 2024 designated 34 critical minerals; see: [Reports](#) [accessed 21 May 2025].

²⁸⁸ DBT, [UK critical minerals midstream and recycling capability report](#), 2 April 2025.

²⁸⁹ [HC Deb 3 December 2024, c40WH](#).

²⁹⁰ McKinsey & Company, [Battery 2030: Resilient, sustainable, and circular](#), January 2023.

The management of end of life for EV batteries is governed by the following regulations:

- The [Waste Batteries and Accumulators Regulation 2009](#), which was [amended in 2015](#).²⁹¹ These regulations implemented European Directive 2006/66/EC.²⁹² They mandate that at least 50% of the materials within used batteries or accumulators must be recycled, and established the concept of ‘[producer responsibility](#)’²⁹³ such that the producers are responsible for collecting batteries and disposing of them.
- [End-of-life vehicles \(ELVs\) regulations](#). These regulations place requirements on manufacturers of all cars and small vans to limit the environmental impact of their disposal.²⁹⁴
- [Waste Electrical and Electronic Equipment \(WEEE\)](#). These regulations aim to reduce the amount of waste from electrical equipment by encouraging the recovery, reuse and recycling of products and components.²⁹⁵

The EU introduced new [Batteries Regulation](#) in August 2023 that aims to ensure that, in the future, batteries have a low carbon footprint, use minimal harmful substances, need less raw materials from non-EU countries, and are collected, reused and recycled to a high degree in Europe.²⁹⁶ UK companies that sell to EU market will need to comply.

Re-using EV batteries

Once they have completed a “first-life” in an EV, batteries can be reused in various ways. These can include grid storage systems,²⁹⁷ and home energy storage for solar panels.²⁹⁸

However, there are limits to reuse options. The House of Lords Environment and Climate Change Committee, in its 2024 EV inquiry, noted that the reuse of batteries in domestic or industrial storage sectors is limited by the fact that

²⁹¹ [Waste Batteries and Accumulators Regulation 2009; The Waste Batteries and Accumulators \(Amendment\) Regulations 2015](#)

²⁹² European Commission, [Directive 2006/66/EC of the European Parliament and of the Council of 6 September 2006 on batteries and accumulators and waste batteries and accumulators and repealing Directive 91/157/EEC](#)

²⁹³ Environment Agency, [Waste batteries: producer responsibility](#), 20 March 2023

²⁹⁴ Office for Product Safety and Standards and DEFRA, [Regulations: end-of-life vehicles](#), 1 January 2021

²⁹⁵ Office for Product Safety and Standards and DEFRA, [Regulations: Waste Electrical and Electronic Equipment \(WEEE\)](#), 11 December 2023

²⁹⁶ European Commission, [Circular economy: New law on more sustainable, circular and safe batteries enters into force](#), August 2023

²⁹⁷ Nissan, [Europe’s largest energy storage system is now live at the Johan Cruyff Arena](#), 29 June 2018, accessed on 29 April 2021

²⁹⁸ Powervault, [Powervault and Renault give EV batteries a “second-life” in smart energy deal](#), 5 June 2017, accessed: [29 April 2021]

EV batteries are currently regulated, certified, and approved for use only in cars.²⁹⁹

The government's response to the committee report set out an intention to consult on reforms to the UK Batteries Regulations in 2024,³⁰⁰ but this has yet to be published.

Re-cycling EV batteries

Recycling of EV batteries is expected to be a growing sector. The IEA note in its [Outlook for battery and energy demand](#) (April 2024) that:

The battery recycling sector, still nascent in 2023, will be core to the future of EV supply chains, and to maximising the environmental benefits of batteries. [...] If all announced projects are developed in full and on time, global recycling capacity could exceed 1 500 GWh in 2030, of which 70% is in China, and about 10% each in Europe and the United States.³⁰¹

In the UK, the [first battery recycling facility](#) was opened by the Recyclus Group in July 2023 in Wolverhampton. It is licensed to recycle up to 22,000 tonnes of Li-ion cells a year.³⁰² There is also extensive research for the battery sector which includes recycling. This includes UK Research and [Innovation's Faraday battery challenge](#), with £610 million of funding until 2025 to drive strong battery business in the UK, and the Automotive Transformation Fund (ATF), operated by the UK Advanced Propulsion Centre which aims to establish a competitive and sustainable UK supply chain with a share of up to £850 million of funding.³⁰³

In addition to other parts of the supply chain, the ATF has supported companies like Altilium who are working to develop new UK battery recycling capabilities to extract raw materials that can then re-enter the supply chain.³⁰⁴

Despite this support for battery recycling, the House of Lords Environment and Climate Change Committee concluded in their 2024 report that:

The UK is not maximising the recycling of EV batteries. The UK also has insufficient capacity to recover critical materials from batteries and retain them in the UK.³⁰⁵

²⁹⁹ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, para 333.

³⁰⁰ [Government response to Lords Environment and Climate Change Committee Report 2024](#), April 2024 (PDF), Recommendations 48-51.

³⁰¹ IEA, [Outlook for battery and energy demand](#), April 2024.

³⁰² Recyclus group, [The UK's first battery recycling plant opens](#), 21 July 2023 .

³⁰³ UKRI, [Faraday battery challenge](#), 7 June 2024; Advanced propulsion centre UK, [Automotive transformation fund](#) (accessed 10 July 2024).

³⁰⁴ APC, [ATF Feasibility study 3 Winners](#) (accessed 10 July 2024).

³⁰⁵ Lords Environment and Climate Change Committee, [EV strategy: rapid recharge needed](#), 6 February 2024, HL Paper 51 2023-24, para 352.

As above, the government response noted the Committee's recommendations and set out an intention to consult in 2024.³⁰⁶

³⁰⁶ [Government response to Lords Environment and Climate Change Committee Report 2024](#), April 2024 (PDF).

5

Past government EV policy pre-2020

Since the late 2000s, successive UK governments have introduced different strategies to assist the move to low emission, and later zero-emission, vehicles, with the Theresa May government committing to end the sale of new conventional petrol and diesel cars and vans by 2040.

5.1

Labour government 1997-2010

The Labour government published its ultra low emission strategy in April 2009.³⁰⁷ It said it would provide £20 million “seed money” to support the development of lead cities and regions in building the necessary charging infrastructure to help increase consumer confidence that would make “ultra-low carbon vehicles” viable. The strategy also expected the private sector ultimately to take the lead in infrastructure provision.

The Labour manifesto for the 2010 General Election promised to “ensure there are 100,000 electric vehicle charging points by the end of the next Parliament.”³⁰⁸

5.2

Coalition government 2010-2015

The 2010 [Coalition Agreement](#) contained a commitment to “mandate a national recharging network for electric and plug-in hybrid vehicles”.³⁰⁹ In delivering on this mandate, the government’s June 2011 [electric vehicles infrastructure strategy](#) said that its approach was “not to mandate ‘a chargepoint on every corner’ – this is not necessary to help the market grow and would be uneconomic”.³¹⁰ Rather, it said the majority of recharging is likely to take place at home and at work, so an extensive public recharging infrastructure would be underutilised and uneconomic. Labour said at the time that this represented a renege on the Coalition’s commitment to a ‘national charging network’.³¹¹ However, others, including manufacturers of electric vehicles, supported the government’s claim that most charging would

³⁰⁷ DfT, [Ultra Low Carbon Vehicles in the UK](#) [PDF], April 2009, p8

³⁰⁸ Labour Party, [A Future fair for All: the Labour Party Manifesto 2010](#) [PDF], April 2010, p1.8

³⁰⁹ HMG, [The Coalition: our programme for government](#) [PDF], May 2010, p31

³¹⁰ HMG [Making the Connection: The Plug-In Vehicle Infrastructure Strategy](#) [PDF], executive summary.

³¹¹ Labour Party press notice, “[Ministers must come clean over attempt to bury bad news on strike day – Woodcock](#)”, 1 July 2011; also reported in: “[Coalition scraps national network of charging points for electric cars](#)”, The Independent, 2 July 2011

be done at home or in the workplace and that the need for public recharging points was therefore limited.³¹²

The government's April [2014 strategy paper on ultra low emission vehicles](#) pledged that by the end of 2014 there would be a rapid chargepoint at every motorway service station and that there would be a network of over 500 rapid chargers across the country by March 2015. It also pledged £32m for charging infrastructure in 2015-20.³¹³

In 2014, the Office for Zero Emission Vehicles announced that funding of £40 million would be made available for the Go Ultra Low Cities scheme. Eight 'cities' were funded to create exemplars for reducing carbon emissions, improving air quality and promoting electric vehicles: Bristol (£7 million), London (£13 million), Milton Keynes (£9 million), Nottinghamshire and Derby (£6 million), with the remaining £5 million split across Dundee, the North East, Oxford, and York.³¹⁴

5.3 Conservative government 2017-2019

The Conservative government published its [Road to Zero Strategy](#) in 2018. This strategy outlined how it would support the transition to zero emission road transport and reduce emissions from conventional vehicles during the transition. The strategy was "long term in scope and ambition, considering the drivers of change, opportunities and risks out to 2050 and beyond".³¹⁵ It set out several measures, including plans to phase out the sale of petrol and diesel vehicles and the rollout charging infrastructure.³¹⁶

5.4 Select Committee reports on electric vehicle policy, 2010-16

The Transport Select Committee published a report on ultra low emission vehicles in September 2012: [Low Carbon Vehicles](#) with the [Government's response](#) published in January 2013.³¹⁷ The Environmental Audit Select Committee assessed policies for ULEVs in its September 2016 report:

³¹² See, e.g. comments by Nissan in "Hammond criticised over car charging points", Financial Times, 1 July 2011.

³¹³ OZEV, [Investing in ultra low emission vehicles in the UK, 2015 to 2020](#) [PDF], April 2014, p16

³¹⁴ DfT & OZEV, [£40 million to drive green car revolution across UK cities](#), 25 Jan 2016

³¹⁵ DfT, [Reducing emissions from road transport: Road to Zero Strategy](#), July 2018

³¹⁶ DfT, [Reducing emissions from road transport: Road to Zero Strategy](#), July 2018

³¹⁷ Transport Committee, [Low Carbon Vehicles](#) [PDF], fourth report of session 2012-13, HC 239, 20 September 2012; [Government Response to the Committee's Fourth Report of Session 2012-13](#) [PDF], eighth special report of 2012-13, HC 884, 21 January 2013.

[Sustainability at the DfT](#).³¹⁸ The [Government response](#) was published on 11 November 2016.³¹⁹

³¹⁸ EAC, [Sustainability in the Department for Transport](#) [PDF], (Third Report of Session 2016–17), HC 184, 1 September 2016, para 25, p14; the response is available at: [Fourth Special Report of Session 2016–17](#), HC 819, 11 November 2016.

³¹⁹ [Government Response Fourth Special Report of Session 2016–17](#) [PDF], HC 819, 11 November 2016.

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